

UNIT -I
ELEMENTARY CONCEPTS

Elementary Concepts:-

DATE

Type of charges

There are three type of charges.

1. Protons — +ive charged particles → Ernest Rutherford
charge — $+1.6727 \times 10^{-19} \text{ C}$ (1919)
Mass — $1.6727 \times 10^{-27} \text{ kg}$

2. Electron — -ive charged particles → J.J. Thomson
charge — $-1.6 \times 10^{-19} \text{ C}$ (19th century)
Mass — $9.11 \times 10^{-31} \text{ kg}$

3. Neutron — neutral charge particles → James Chadwick
Mass — $1.6729 \times 10^{-27} \text{ kg}$. (1932)

Mass of neutron & electrons Protons are same —
and they are 1840 times heavier than Electrons.

Atoms:-

An atom is the smallest constituent of matter that has the properties of chemical elements.

Every solid, liquid, gas & plasma comprises of neutral or ionized atoms.

Structure of Atom:-

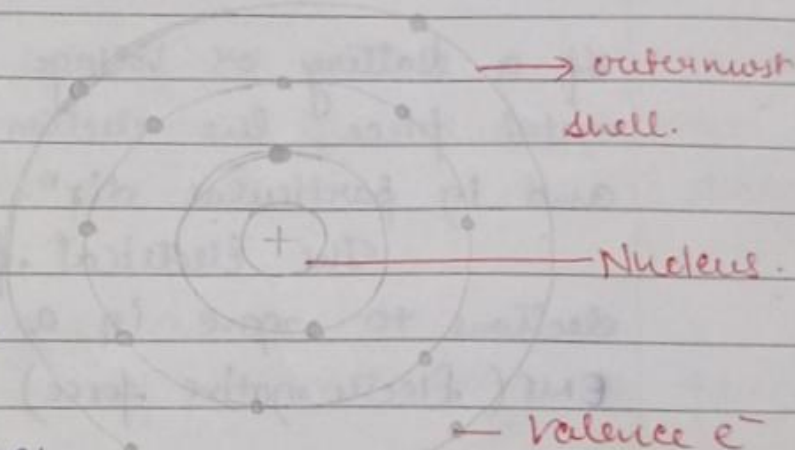
Nucleus — comprises of proton & neutrons.

No. of protons = no. of neutrons

No. of proton — atomic no.

no. of proton + Neutron — atomic mass.

classmate



PAGE

if no. of proton = no. of neutron $\therefore$ then the atom is electrically neutral.

If atom has more or fewer electrons than proton, then it has +ive or -ive charge, then charged atom is called ion.

$$\text{eg} \rightarrow P = 7 \\ N = 3$$

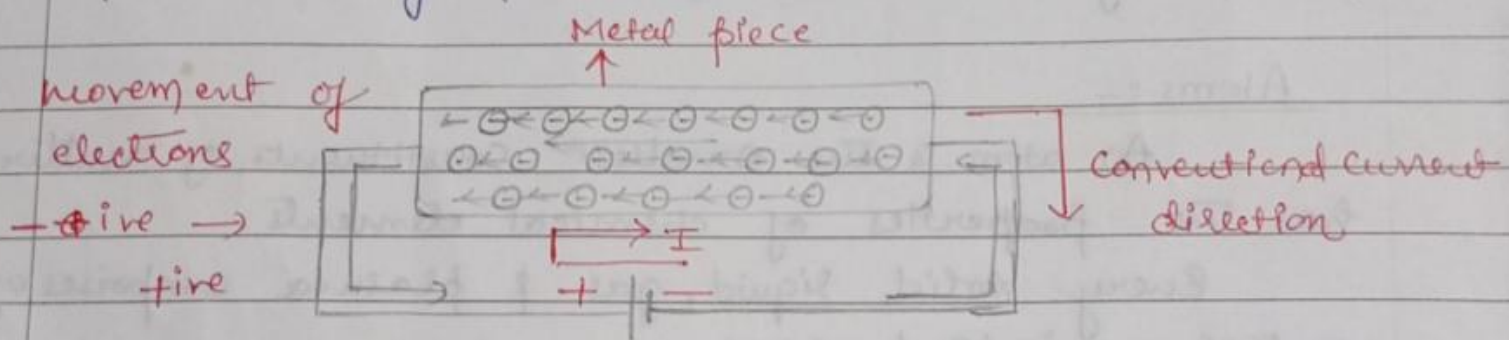
$$\therefore P - N = 7 - 3 = 4.$$

Concept of EMF & Current :-

Concept of Emf :-

EMF - Electromotive force

All metal contains a large no. of free electrons, which move randomly in all the dirⁿ, inside the metal piece.



If a battery or voltage source is connected a/c the metal piece, the electron starts moving from -ive to +ive and in particular dirⁿ.

The Electrical force or pressure that causes the electrons to move in a particular dirⁿ is called EMF (Electromotive force)

Unit of EMF - Volts.

Unit of current (I) - Amperes.

Election Current -

The current due to the flow of e^- is called electron current.

negative
end of one
battery

Electron flow Current

the end of
the
battery.

Conventional Current -

The current is said to be flowing from +ve end (higher potential) to -ve end (lower potential) & it is said that it flows opposite to the electron current.

Conventional Current

Electron Current

Relationship b/w Current & Charge-

Current is defined as the flow of electron.
Higher the flow of electrons, Higher the value of current.

Each electron carries current of $-1.6 \times 10^{-19} \text{ C}$ charge

Hence,

Current can also be defined as -

'The rate of flow of electrons' charges in per unit time'

DATE

mathematically defined as,

$$\text{current} = \frac{\text{flow of charge (Q)}}{\text{time (t)}}$$

$$I = \frac{Q}{t}$$

$$\therefore Q = It$$

Definition of 1 amp -

A current of one ampere is said to be ...

"If a charge of one coulomb passes thr. any given point in one second is called One ampere of current"

$$I = \frac{Q}{t}$$

$$I = \frac{1 \text{ Coulomb}}{1 \text{ Sec}}$$

$$\text{charge of } e^- = 1.6 \times 10^{-19} \text{ C}$$

$$1 \text{ C} = \frac{1}{1.6 \times 10^{-19}} e^-$$

$$n_e = 6.25 \times 10^{18} e^-$$

$$I = 6.25 \times 10^{18} e^- / \text{Sec.}$$

$\therefore$ To produce 1 amp. of current, $6.25 \times 10^{18} e^-$ passes any given pt. in one second.

Electric Potential →

When a body is charged, work is done in charging it, this work done is stored in the form of potential energy.

The charged body has the capacity to do work by moving other charges either by attraction or repulsion. This ability of one charged body is called Electric potential.

The capacity of one charged body to do work is called Electric Potential.

$$\text{Electric potential (V)} = \frac{\text{Work done } \left[\frac{(W)}{Q} \right]}{\text{charge}}$$

$$= \frac{\text{Joules}}{\text{Coulomb}} = \text{Jc}^{-1} \text{ (or Volt)}$$

One volt —

Hence, a body is said to have '1 volt' if 1 joules of work ^{is} done to give it a charge of 1 coulomb.

$$\therefore 1 \text{ V} \xrightarrow{\text{to give}} \text{ie } 1 \text{ Joules} \rightarrow 1 \text{ C}$$

Potential Difference —

The difference in the potentials of two charged bodies is called Electric Potential Difference.

A → 5V → 5 Joules | — work is done to give it a
B → 3V → 3 Joule | — charge of 1C.

+5V

+3V

(A)

(B)

Higher
potentiallower
potential

Electrons will flow from →

B → A

* when when two bodies attain the same potential the current will stop flowing.

Thus —

No potential difference



No current flow.

∴ To flow the current the a circuit the potential difference must exist.

Resistance :-

Resistance is defined as the 'opposition of the flow of current' measured in Ohm (Ω)

* Resistance of metal is small (lower) - good conductor of electricity.

* Materials like, wood, plastic, glass - do not allow the current to flow through it. It is called a bad conductor of electricity or insulator.

Mathematical expression -

$$R = \frac{\rho l}{a}$$

a - area of cross-section (m^2)

l - length of the conductor (m)

ρ - resistivity of the material & constant.

Factors affecting the resistance value -

The factors affecting the value of resistance are -

1. Length of the conductor -

$$R \propto l$$

Resistance is directly proportional to length, as the length of a conductor increases, its resistance also increases and vice-versa.

2. Area of cross-section -

Resistance is inversely proportional to the area of cross-section of the condⁿ.

ie $R \propto \frac{1}{a}$

ie if area of cross-section of condⁿ ↑, its resistance decreases and vice-versa.

Thick condⁿ has large cross-section area, ∴ it has ^{lower} higher resistance.

whereas, thin condⁿ has small cross-section area ∴, has higher resistance.

Hence, thick condⁿ



large cross-section
area



low resistance

thin condⁿ



small cross-section
area



high resistance.

3. Type of material —

The value of resistance depends on the type of material.

$R \propto \rho$

Good condⁿ — low resistance

Semi-condⁿ — Medium resistance

Insulator — High resistance.

4. Temperature —

Due to the change in temp. the length of the condⁿ changes, ∴ the value of resistance changes.

ok, the $\uparrow$ in temp. $\longrightarrow$ length of one condⁿ $\uparrow$ ses
results in the $\uparrow$ se $\longleftarrow$ |
in the resistance of condⁿ.

Resistivity:

Ity:
— also called as specific resistance and denoted by ρ (p) — ohm.

$$R = f \frac{l}{a}$$

$$\therefore P = \frac{q}{l} \cdot R$$

$$f = \frac{m^2 \times 2}{m} = 2 \times m$$

$p = 2.7 \text{ m} \Rightarrow \text{mpho.}$

The specific resistance or resistivity of one material is defined as the resistance of a piece of condⁿ of 1 mt. length and of 1mm^2 cross-section area.

$$\text{ie } R = \frac{1}{1} \times R$$

$$Q = 1 \text{ m}$$

$$a = 1 \text{ m}^2$$

$$\therefore f = R$$

Best Condⁿ — Silver — low value of resistance or specific res
Best insular ————— high —————

Defination of 1-2 resistance -

The resistance of the circuit is said to be 1- Ω , if the current flowing of 1 amp. generates the heat at one rate of one joule per second.

$$H = I^2 R t$$

ie $I = 1 \text{ Amp.}$

$R = 1 \Omega$.

$t = \text{One second}$

$$\therefore H = (1)^2 \times 1 \times 1$$

$$H = 1 \text{ joule.}$$

$$1 \text{ joule} = 0.24 \text{ calories.}$$

Conductance:-

It is defined as the reciprocal of resistance. Also defined as —

'The degree to which an object conducts electricity'

G — conductance

$$G = \frac{1}{\text{Resistance}}$$

$$= \frac{1}{2} = 0.5 \text{ — (mho) or Siemens}$$

Conductivity:-

The conductivity is property of one material, which allows the current to flow through it.

— or —

Electrical Condⁿ is a measure of the material ability to conduct electric current.

denoted by (σ) —

and also,

defined as the reciprocal of resistivity.

σ ρ

$$\left[\sigma = \frac{1}{\rho} \right]$$

$$\sigma = \frac{1}{2 \cdot m} = 28 \cdot m^{-1}$$

$$\left[\sigma = \text{mho/meter} \cdot \text{or} \cdot \text{seimens/m} \right]$$

Que/- The resistivity of Copper is $1.724 \mu\Omega\cdot m$. find the resistance of a copper rod having diameter 1mm & length 1m .

Solⁿ $\rightarrow$

$$R = \rho \frac{l}{a}$$

$$l = 1\text{m}$$

$$a = \pi r^2 \rightarrow \pi \left(\frac{d}{2} \right)^2 = \pi \times \frac{1 \times 10^{-3} \times m}{4}$$

$$R = \frac{1.724 \times 10^{-8} \Omega\cdot m \times 1\text{m}}{\pi \times \frac{(10^{-3})^2}{4} \text{m}}$$

$$R = 2.196 \times 10^{-3} \Omega\cdot m$$

$$= \underline{\underline{0.002196 \Omega}}$$

Que/- find the resistance of one following annealed Copper wires-

1. 1mm^2 cross-section, 100m long

2. 25cm^2 cross-section, 200m long.

Given that ρ is $1.73 \mu\Omega\cdot m$.

Solⁿ $\rightarrow \rho = 1.73 \mu\Omega\cdot m$ — given.

$$(i) \quad R = \rho \frac{l}{a} \rightarrow \frac{1.73 \times 10^{-8} \Omega\cdot m \times 100}{(1000)^2}$$

$$[1.73 \times 10^{-6} \times 1]$$

$$1\text{m} = 100\text{cm} = 10^2\text{cm}$$

$$1\text{m} = 1000\text{mm}$$

$$1\text{mm} = \frac{1}{1000}\text{m}$$

$$\frac{1\text{m}}{10} = 0.1$$

$$(i) R = \rho \frac{l}{a} \Rightarrow 1.73 \times 10^{-6} \times 2\text{cm} \times 100\text{m} \left(\frac{1}{1000}\right)^2$$

$$\Rightarrow \frac{1.73 \times 10^{-6} \times 2\text{cm} \times 10^6 \times 100\text{m} \times 10^2}{10^2}$$

$$= \underline{1.73 \times 2\text{cm}}$$

$$(ii) R = \rho \frac{l}{a}$$

$$\Rightarrow \frac{1.73 \times 10^{-6} \times 100 \times 200}{25 \times 10^{-2}}$$

$$= \frac{1.73 \times 10^{-6} \times 100 \times 200 \times 10^2 \times 10^2}{25}$$

$$R = \underline{0.13842 \times 10^2}$$

$$(ii) R = \rho \frac{l}{a}$$

$$\Rightarrow \frac{1.73 \times 10^{-6} \times 200 \times 10^2}{25}$$

$$= 0.1384 \times 10^{-2}$$

$$= \underline{0.0013842}$$

Ques- Calculate the resistance of a 200m length of wire having a uniform cross-sectional area of 0.01mm^2 & resistivity of $50\mu\Omega\text{-cm}$. If wire is drawn out four times

of its original length. Calculate resistance.

Solⁿ $l = 200\text{m}$ $a = 0.01\text{mm}^2$ $\rho = 50 \times 10^{-6} \Omega\text{-cm}$.

Original resistance $\rightarrow$

$$R = \rho \frac{l}{a}$$

$$R = 50 \times 10^{-6} \Omega\text{-cm} \times \frac{200 \times 100 \text{ cm}}{0.01 \times 10^{-2} \text{ cm}^2}$$

$$R = \frac{5000 \Omega\text{-cm} \times 20}{0.01} = \underline{\underline{10,000 \Omega}}$$

if length increase by 4-times,

then resistance also increases by 4 times of its original resistance

also area reduce by 4 times
ie $a/4$

$$R_1 = \frac{\rho \times 4l}{a/4}$$

and original —

$$R = \frac{\rho \times l}{a}$$

$$\frac{R_1}{R} = \frac{\frac{\rho \cdot 4l}{a/4}}{\frac{\rho \cdot l}{a}}$$

$$\underline{\underline{R_1 = 16R}}$$

Volume $\rightarrow a \times l$

$$a \times l = a' \times l' \quad \text{(original) (New)}$$

$$a' = \frac{a \times l}{l'}$$

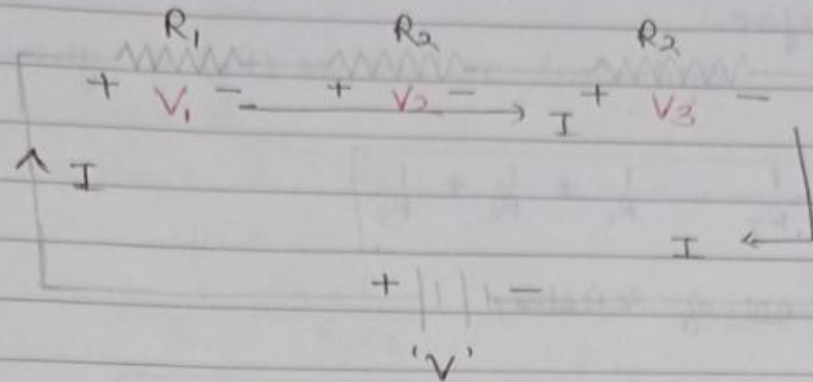
$$= \frac{a \times l}{4l}$$

$$\boxed{a' = a/4}$$

$$\therefore R_1 = 16 \times 10,000 = \underline{\underline{160,000 \Omega}} = \underline{\underline{160 \text{ K}\Omega}}$$

Resistance in series -

Fig shows of one three resistance are connected in series with each other across a supply voltage 'V'.



- All the elements connected in series carry the same current.
- But, the voltage a/c individual elements (if connected in series) varies, i.e. V_1, V_2, V_3 — called as voltage drop.

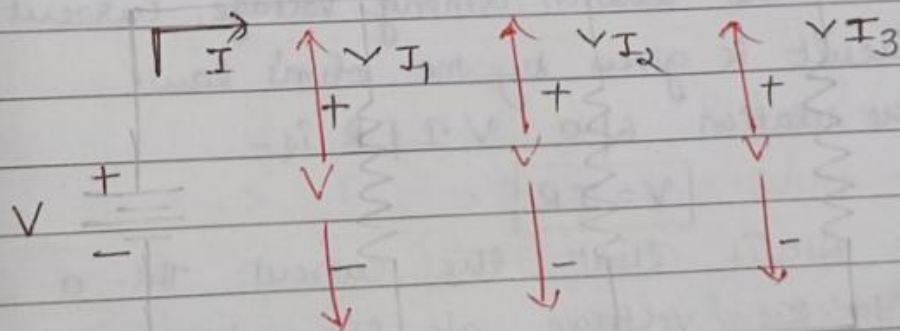
$$\therefore \text{Eq}^n \text{ resistance } (R_T) = R_1 + R_2 + R_3$$

$\therefore$ for 'n' resistor connected in series -

$$\boxed{\text{Eq}^n \text{ resistance } (R_T) = R_1 + R_2 + R_3 + \dots + R_n}$$

Resistance Connected in parallel -

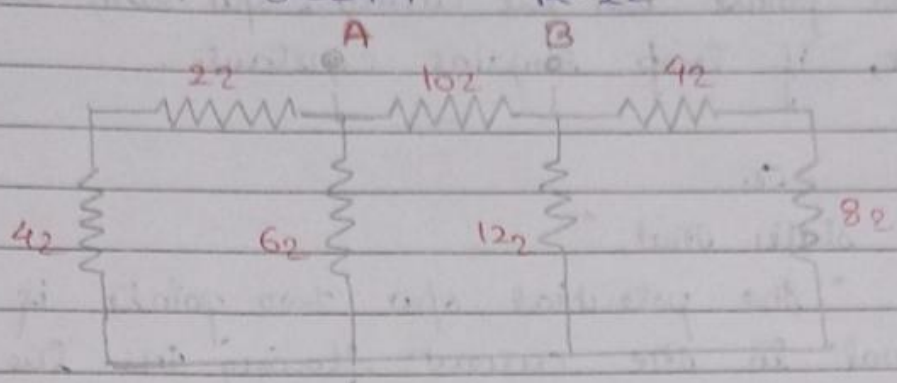
Fig shows of the resistance connected in parallel a/c one DC voltage supply.



$$R_{eq} = \frac{1}{\frac{1}{R_{eq}}} = \frac{1}{\frac{1}{8.222} + \frac{1}{4}}$$

$$R_{eq} = \frac{8.22 \times 4}{8.22 + 4} = \frac{32.88}{12.22} = \underline{\underline{2.692}}$$

Que.

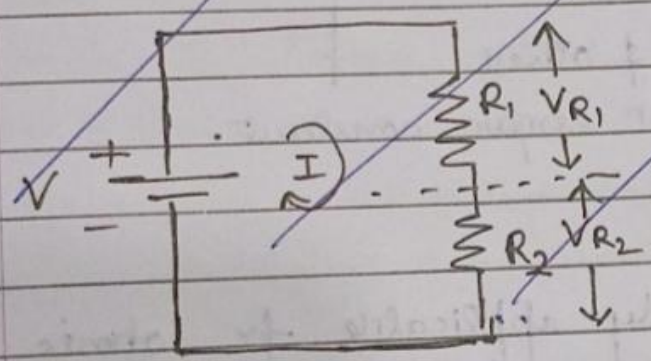


Calculate R_{AB}

$$R_{AB} = \underline{\underline{5.452}}$$

Current Divider Rule:-

Division of Voltage in the series resistor $\rightarrow$



$$V = V_{R1} + V_{R2}$$

$$V_{R1} = \frac{V \times R_1}{(R_1 + R_2)}$$

$$R = R_1 + R_2$$

- In parallel circuit, Voltage remains constant (or same) of every individual branch if connected in parallel.
- where as the current gets divided.

Therefore -

equivalent resistance (R_T) is given as

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

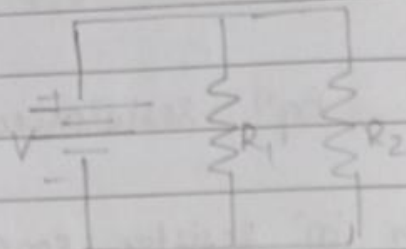
for 'n' no. of resistor,

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

Effective resistance of two resistor in parallel -

$$\therefore \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_T} = \frac{R_1 + R_2}{R_1 R_2}$$



$$\therefore R_T = \frac{R_1 R_2}{R_1 + R_2}$$

Ohm's Law:-

The relation among Voltage, current & resistance in a circuit is given by ohm's law.

The relation b/w V, I & R is -

$$V = IR$$

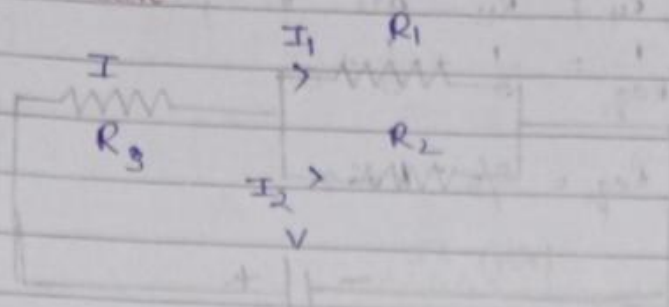
Ohm's law states that the current thru a Condⁿ b/w two points directly $\propto$ to the voltage a/c the two points.

classmate

$$I \propto V$$

$$R = \frac{V}{I}$$

Series parallel Circuit →



R_1 & R_2 in parallel
 $\therefore \frac{1}{R_{eq1}} = \frac{1}{R_1} + \frac{1}{R_2}$

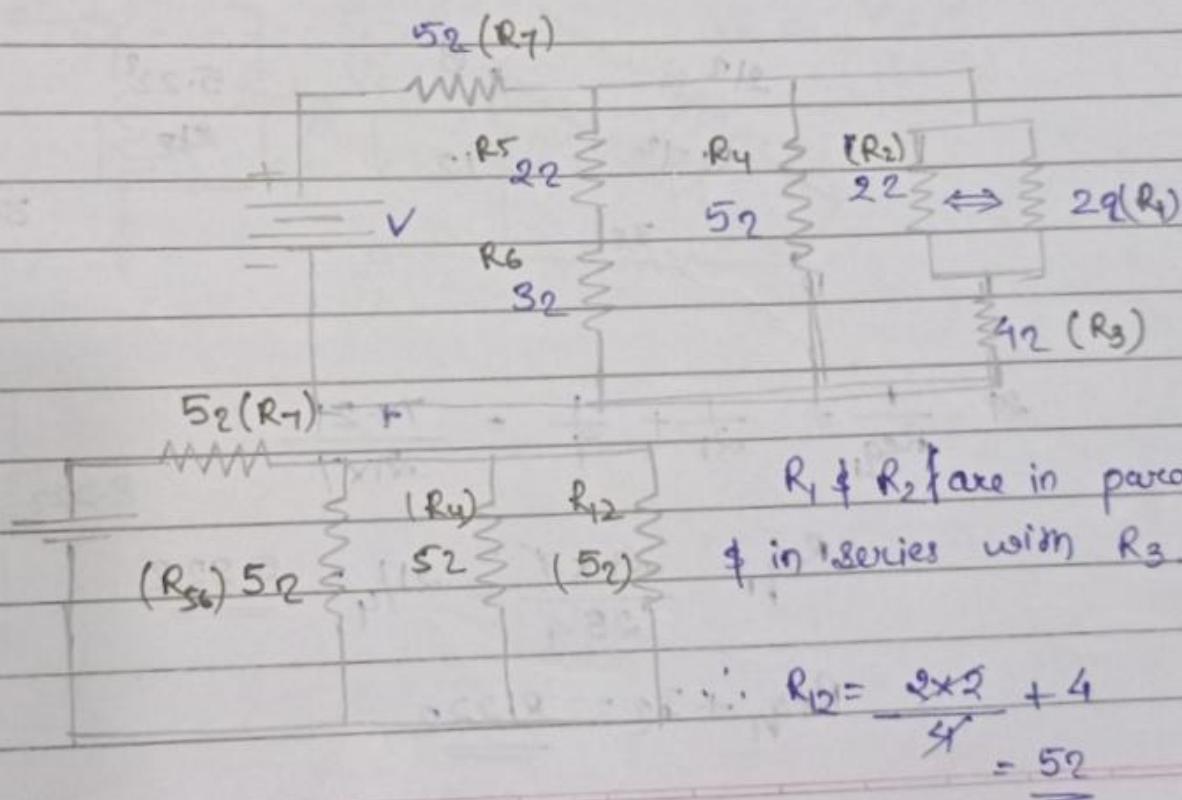
$$= \frac{1}{R_{eq1}} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R_{eq1} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{eq} = R_3 + R_{eq1}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

Ex - Simplify the circuit -



R_1 & R_2 are in parallel
 & in series with R_3

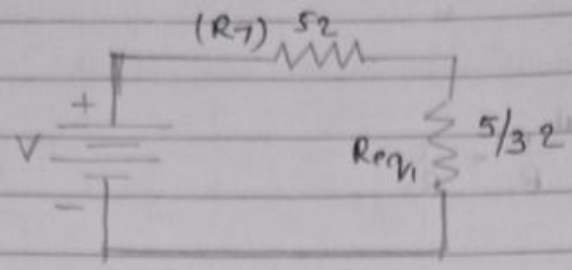
$$\therefore R_{12} = \frac{2 \times 2}{4} + 4$$

$$= 5\Omega$$

$R_{S6}, R_4 \text{ \& } R_{12}$ are in parallel with each other.

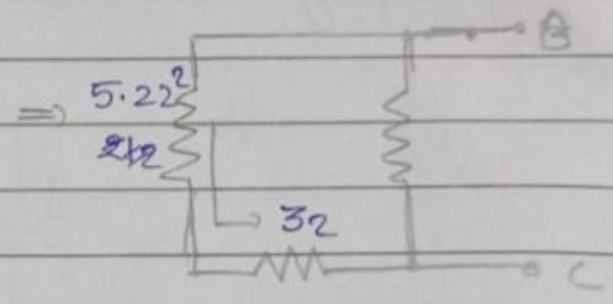
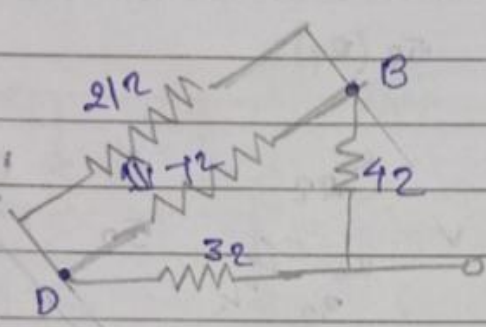
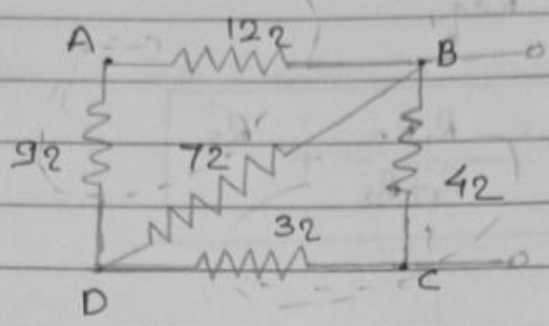
$$\therefore \frac{1}{R_{eq1}} = \frac{1}{5} + \frac{1}{5} + \frac{1}{5}$$

$$R_{eq1} = 5/3 = 1.662$$



$$R_{eq} = 5 + 5/3 = 20/3 = \underline{\underline{6.662}}$$

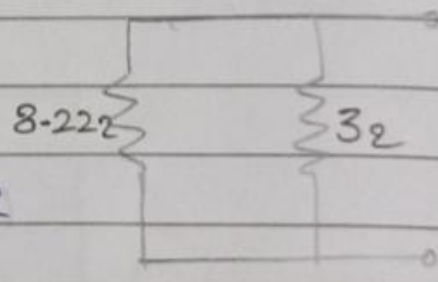
Que - find eqⁿ resistance of -



$$\therefore \frac{1}{R_{eq1}} = \frac{1}{21} + \frac{1}{7} = \frac{7+21}{21 \times 7}$$

$$R_{eq1} = \frac{21 \times 7}{28} = 21/4 = \underline{\underline{5.222}}$$

$$R_{eq1} + 32 = \underline{\underline{8.222}}$$



Ohm's Law —

Ohm's law states that —
"the ratio of potential difference (V) b/w any of the two points and the current ^{flowing} b/w them should be constant, if temp remains constant."

or.

"It states that."

"the potential b/w two points is equal directly proportional to the current flowing thr. the circuit."

ie — $V \propto I$

$$\boxed{V = IR} \quad \text{ohms.}$$

Volt

Limitation —

Ohm's law is fundamental, but it applies under specific condⁿ —

1) physical condⁿ — the temp & other physical properties must remain constant.

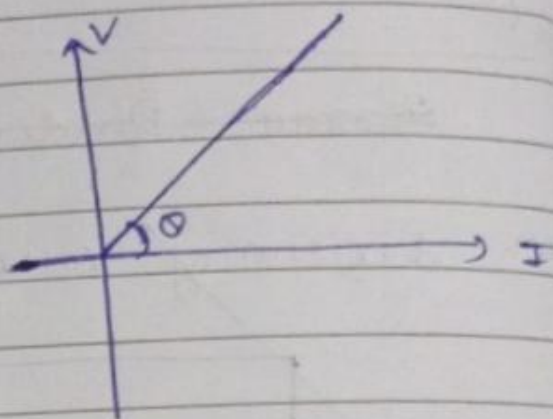
2) Non-ohmic component —

Ohm's law is only applicable for ohmic components, non-ohmic components (like — diode, transistor, bulb filaments) — does not follow ohm's law.

as current flows $\rightarrow$ filament temp $\uparrow$ res $\rightarrow$
resistance varies $\therefore V \& I$ also changes.

$\therefore$ ohm's law is applicable for linear components.

[ohmic components — most metallic condⁿs —
Copper, Silver, aluminium]



DC CIRCUITS

Dc. Circuits

DATE

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Electric Circuit — closed path followed by the current is called as Electric Circuit — Elements — Resistor, Inductor, Capacitor,

(is an electrical confⁿ, which connects various components electrically)
Such as R, L, C , Voltage source & Current source

Dc. Circuits —

The closed path followed by Direct Current (Dc) is called as Dc. circuit.

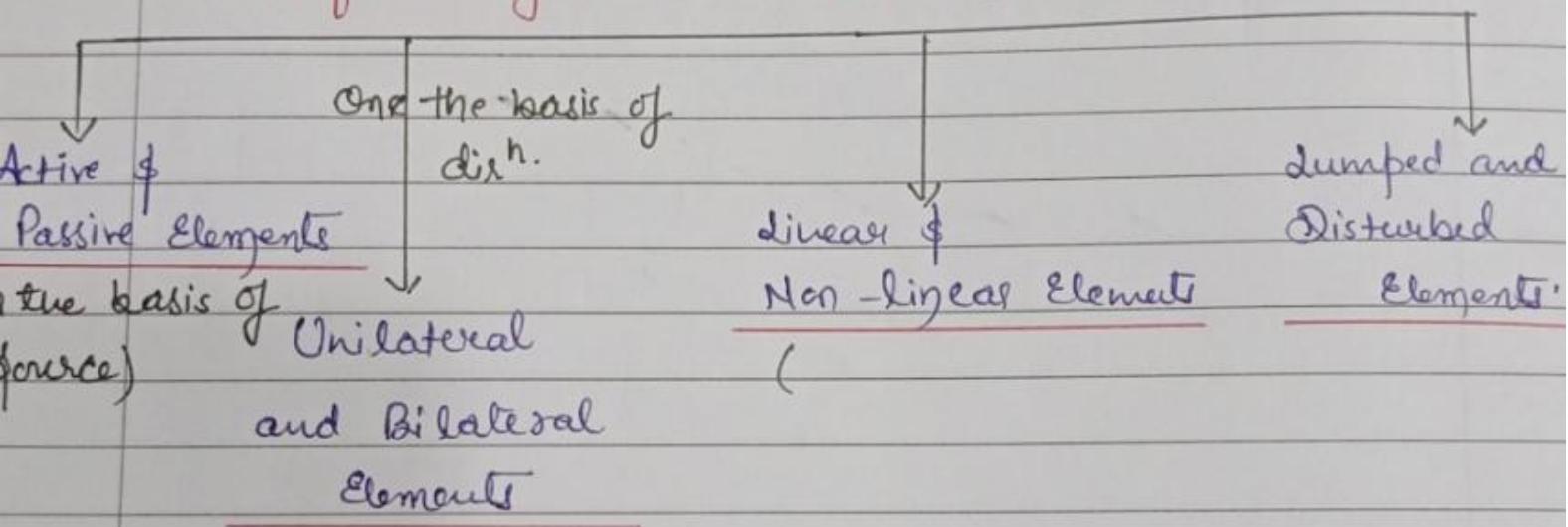
In D.C Circuit loads may i.e. resistance are connected in —

- (i) Series
- (ii) parallel.
- (iii) Series-parallel combination.

Ac Circuit —

The closed path followed by the ~~Alte~~ Alternating current is called as Ac. circuit.

(The circuit elements are categorized as —)
Classification of Electrical Networks.



Active and passive Elements :-

Active Elements — which supply^{ies} energy in the form of voltage or current to one circuit or network.

Eg - current source / voltage source $\textcircled{\uparrow} I$ $\textcircled{\sim} V$
or batteries or generators.

Passive Elements :-

The Elements which receives energy from the source in the form of voltage or current.

Eg - resistor, inductor and capacitor.

Unilateral and Bilateral Elements -

Unilateral Elements -

The Elements which conducts the current in one direction only is called as Unilateral Elements.

Eg - Diode, Transistor, Vacuum tubes etc.

Bilateral Elements -

Elements which conducts the current in both the direction.

Eg - resistor, transmission line.

Linear and Non-Linear Elements -

Linear Elements -

Linear elements are the one whose parameters does not changes with the current or voltage.

ie, constant resistance

— $\sim$ — inductance

— $\sim$ — capacitance

Non-Linear Elements -

Non-linear elements are those that are changed with respect to the voltage and current.

- do not follow - ohm's law.

Eg - diode, transistors and sensors require power to do their job.

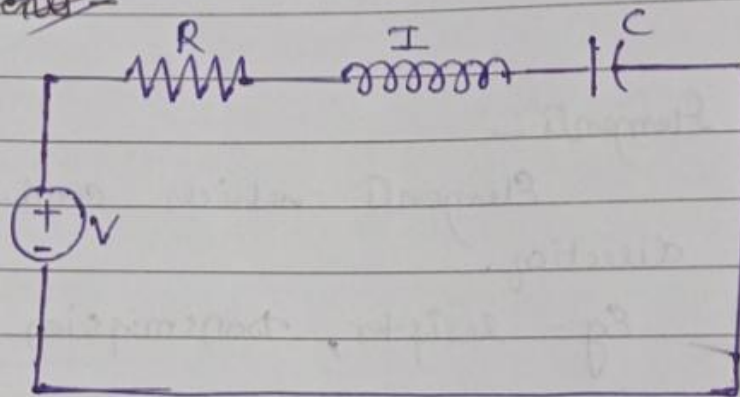
Lumped and Distributed Elements -

Lumped Elements -

A network in which all the elements are physically separable or that are assumed to be concentrated at one place.

are suitable for low frequency analysis, small circuits.

Distributed Elements -



Distributed Network -

A network in which elements are not physically separable, where as they are spread out over a physical length or area,

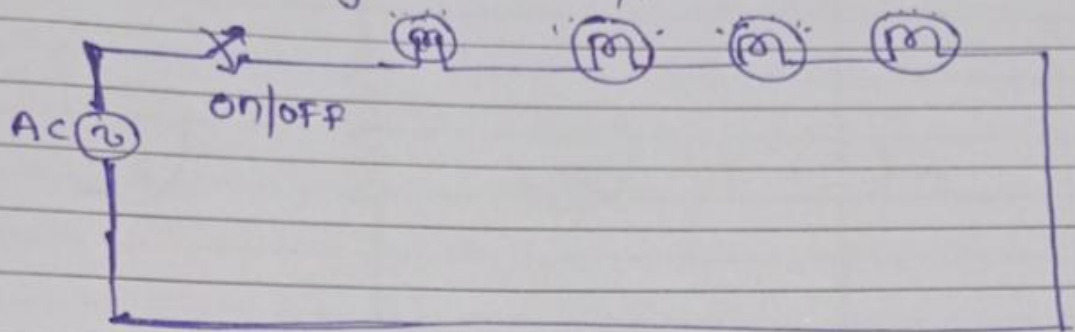
Eg - X-mission line

Circuit Simplification Technique -

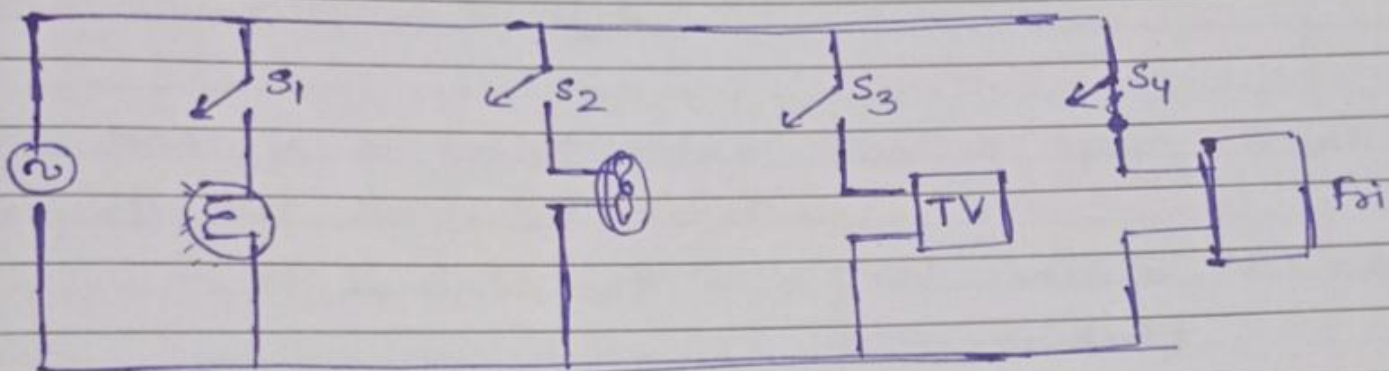
- 1) Series - parallel Combination
- 2) Use redundant branch removal
- 3) Star-delta.

Series-parallel -

- Practical application of series connection -

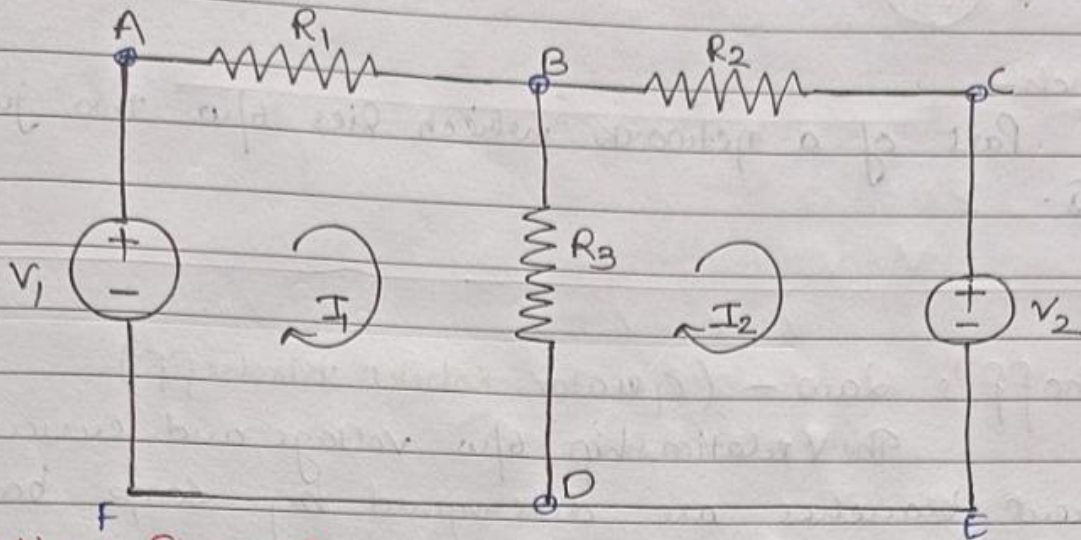


- parallel connection -



Network terminology -

Electrical network is interconnection of electrical components eg -
batteries, inductor, capacitor, resistor.



Active Elements — V_1 & V_2 .

Passive Elements — R_1, R_2, R_3 .

Node — Node is a point where two or more circuit elements are connected together.

— A, B, C, D (It is a point where current gets divided).

Junction —

Junction where three or more circuit elements are connected together.

— B and D.

(It is a point where current gets divided).

Loop —

The closed path of a network.

Eg — ABDA, BCEDB, ABCEDFA.

Mesh-

the elementary form of loop which cannot be further divided.

ABDFA, BCEDB.

Branch-

Part of a network which lies b/w two junction points.

Kirchoff's Law - (Gustav Robert Kirchoff).

The relationship b/w voltage and current of different branches are determined by some basic laws. They are called as Kirchoff's law.

Hence, there are two types of Kirchoff's laws -

- Kirchoff's Current law.
- Kirchoff's Voltage law.

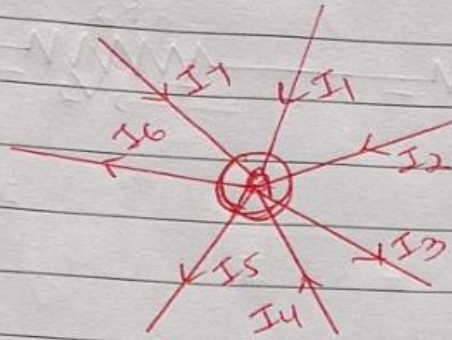
Kirchoff's Current Law - (Nodal Analysis)

KCL - states that -

"The algebraic sum of the electric current meeting at any junction (or node) is zero".

or

"Total no. of current entering the a junction or node are exactly equals to no. of current leaving the node".



$$\boxed{\sum I = 0}$$

Current entering - I_1, I_2, I_4, I_7
(positive sign)

Current leaving - I_3, I_5, I_6
(negative sign)

$$+I_1 + I_2 - I_3 + I_4 - I_5 - I_6 + I_7 = 0$$

$$I_1 + I_2 + I_4 + I_7 = I_3 + I_5 + I_6$$

Kirchoff's Voltage Law - (Mesh Analysis).

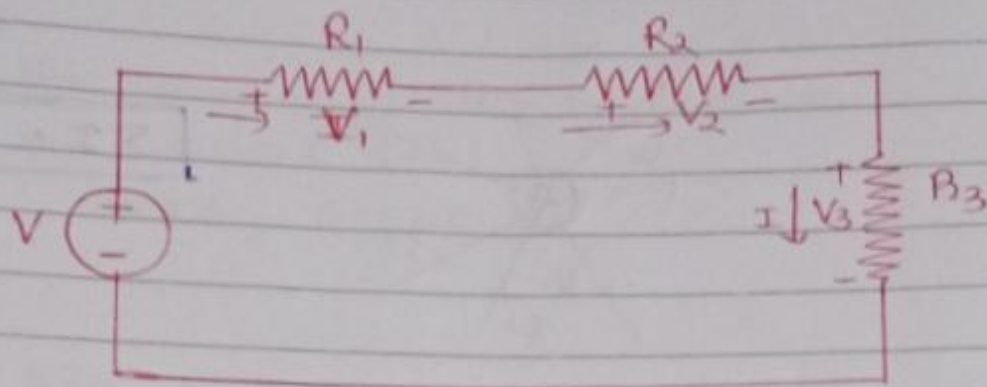
KVL states that -

'In any closed loop network, the total voltage around the loop is equal to the sum of the voltage in the loop drop with in same loops'.

- or -

In any closed circuit, the algebraic sum of the voltage source and voltage drop across resistors are equal to zero.

$$\boxed{\sum V = 0}$$



$$\sum V = 0$$

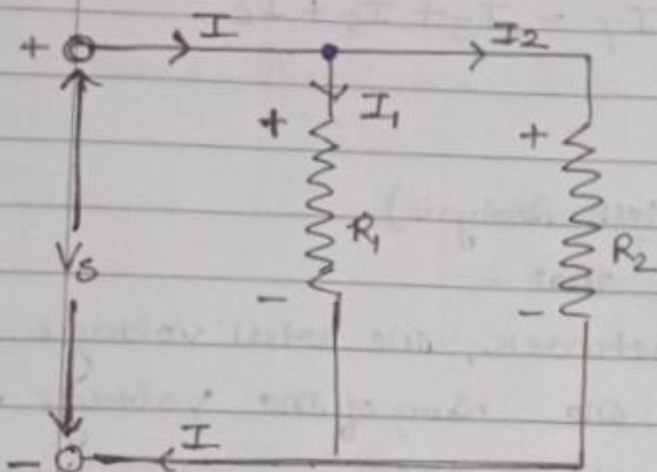
$$V - V_1 - V_2 - V_3 = 0$$

$$V = V_1 + V_2 + V_3$$

$$V = IR_1 + IR_2 + IR_3$$

$$\boxed{V = I(R_1 + R_2 + R_3)}$$

Current Division Rule :- (Parallel connected resistor only).



The main characteristics of the parallel circuit is —

which they may produce different current but the voltage is common to all the connected path.

$$\text{ie } V_{R_1} = V_{R_2} = V$$

$$I = I_1 + I_2$$

$$I_1 = \frac{V}{R_1} \quad I_2 = \frac{V}{R_2}$$

$$I = \left(\frac{V}{R_1} + \frac{V}{R_2} \right)$$

$$I = V \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$I = V \left(\frac{R_1 + R_2}{R_1 R_2} \right)$$

$$V = \left(\frac{R_1 R_2}{R_1 + R_2} \right) \cdot I$$

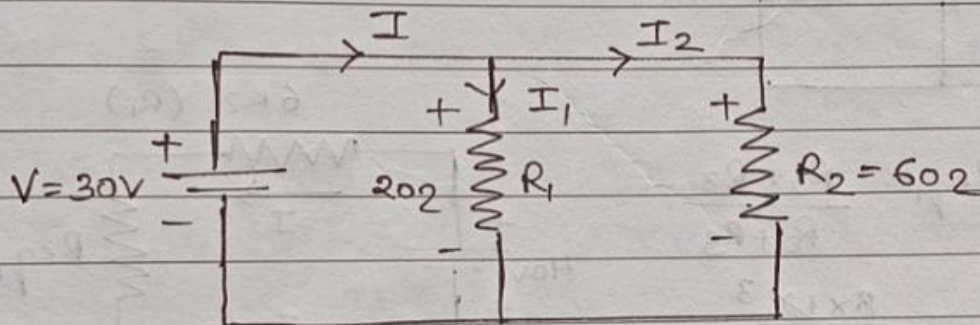
$$I_1 = \frac{V}{R_1} = \left(\frac{R_1 R_2}{R_1 + R_2} \right) \times \frac{1}{R_1} \times I$$

$$I_1 = \frac{R_2}{R_1 + R_2} \times I$$

$$I_2 = \frac{V}{R_2} = \left(\frac{R_1 R_2}{R_1 + R_2} \right) \times \frac{1}{R_2} \times I$$

$$I_2 = \left(\frac{R_1}{R_1 + R_2} \right) \times I$$

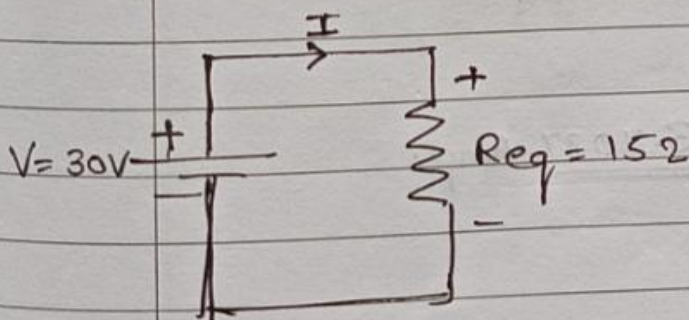
Que. 1 -



Soln. — R_1 & R_2 — are in parallel.

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{eq} = \left(\frac{R_1 R_2}{R_1 + R_2} \right) = \frac{20 \times 60}{20 + 60} = 15\Omega$$



$$\therefore \text{total current } I = \frac{V}{R_{eq}}$$

Current through each resistor $\rightarrow$

I_1 - current through R_1 is -

$$I_1 = \frac{V}{R_1} =$$

Using current divider rule

$$I_1 = \left(\frac{R_2}{R_1 + R_2} \right) \times I$$

$$I_2 = \left(\frac{R_1}{R_1 + R_2} \right) \times I$$

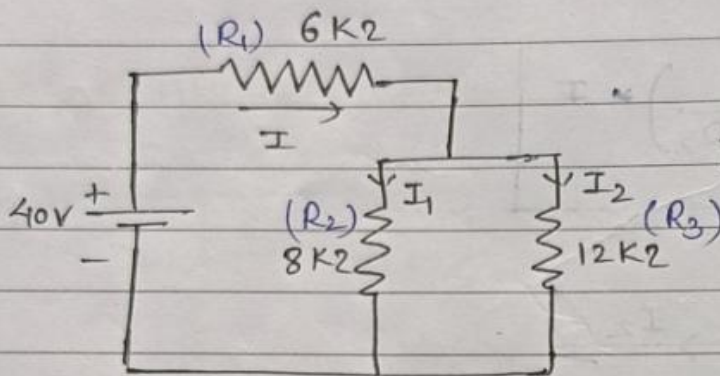
$$I_1 = \frac{60}{80} \times 2$$

$$I_2 = \frac{60}{80} \times 2$$

$$I_1 = \frac{30}{20} = 1.5 \text{ Amp}$$

$$I_2 = 0.5 \text{ Amp}$$

Que 2.

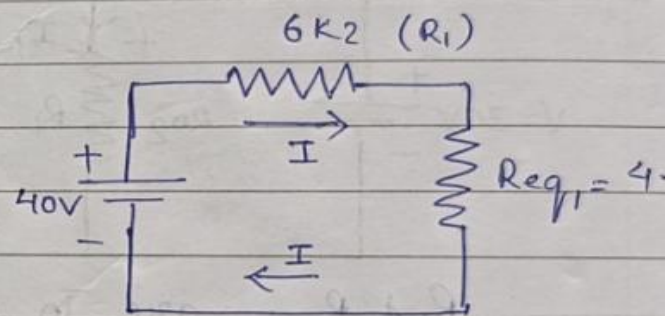


Calculate the current through all the resistances

Soln - $R_2 \parallel R_3 \rightarrow R_{eq1} = \frac{R_2 \times R_3}{R_2 + R_3}$

$$R_{eq1} = \frac{8 \times 12}{8 + 12} = \frac{8 \times 12}{20}$$

$$= \frac{24}{5} = 4.8 \text{ K}\Omega$$



$\therefore R_{eq1} \text{ \& } R_1$ - in series.

$$4.8 + 6 = 10.8 \text{ K}\Omega$$

$$V = IR \Rightarrow I = \frac{40}{10.8} = \underline{\underline{3.7037 \text{ Amp}}}$$

$$I_1 = \left(\frac{R_3}{R_2 + R_3} \right) \times I$$

$$I_1 = \left(\frac{12}{12+8} \right) \times 3.7$$

$$= \underline{\underline{2.22 \text{ Amp}}}$$

$$I_2 = \left(\frac{R_2}{R_2 + R_3} \right) \times I$$

$$I_2 = \left(\frac{8}{12+8} \right) \times 3.7$$

$$= \underline{\underline{1.48 \text{ Amp}}}$$

} current divider rule.

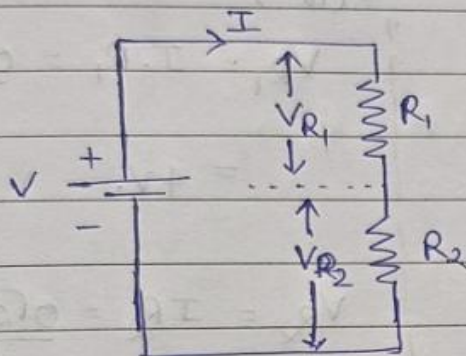
$$\text{or } I = I_1 + I_2$$

$$I_2 = I - I_1$$

$$= 3.7 - 2.22$$

$$= \underline{\underline{1.48 \text{ Amp}}}$$

Voltage Divider Rule -



V_{R1} - Voltage across resistor (R_1)

V_{R2} - (R_2)

In series current remains same.

$$\therefore V = V_{R1} + V_{R2}$$

$$V_{R1} = IR_1 \quad V_{R2} = IR_2$$

$$V = I(R_1 + R_2)$$

$$\boxed{I = \frac{V}{R_1 + R_2}}$$

$$V_{R1} = IR_1$$

$$= R_1 \times \frac{V}{R_1 + R_2}$$

$$\Rightarrow \frac{V_{R1}}{R_1} = \frac{R_1 \times V}{R_1 + R_2}$$

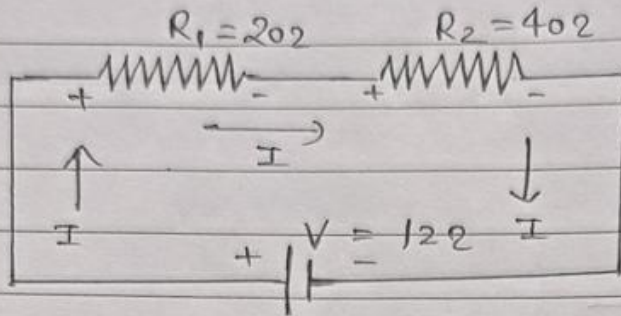
$$\boxed{V_{R1} = \frac{R_1}{R_1 + R_2} \cdot V}$$

$$V_{R_2} = I \cdot R_2$$

$$V_{R_2} = \frac{R_2}{R_1 + R_2} \times V$$

$$V_{R_2} = \left(\frac{R_2}{R_1 + R_2} \right) \times V$$

Que. 1/-



Calculate the voltage across the resistors, by voltage divider rule.

Solⁿ

R_1 series with R_2 -

$$R_{eq} = R_1 + R_2 = 20 + 40 = 60\Omega$$

$$V = IR$$

$$I = \frac{V}{R_{eq}} = \frac{12}{60} = \frac{1}{5} = 0.2 \text{ Amp.}$$

$$V_{R_1} = \left(\frac{R_1}{R_1 + R_2} \right) \times V$$

$$= \frac{20}{60} \times 12$$

$$V_{R_1} = 4V$$

$$V_{R_2} = \left(\frac{R_2}{R_1 + R_2} \right) \times V$$

$$= \frac{40}{60} \times 12$$

$$V_{R_2} = 8V$$

← or →

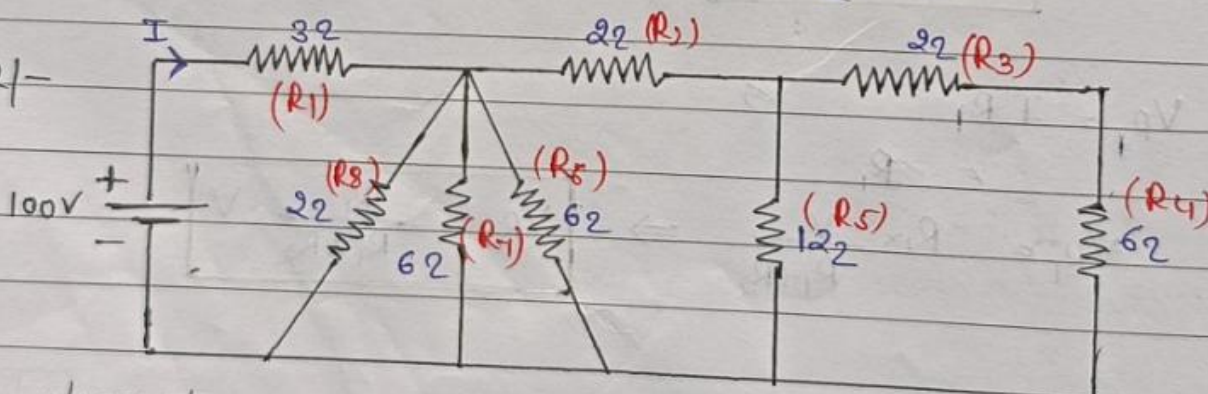
$$V_{R_1} = I \cdot R_1 = \frac{0.2 \times 20}{10}$$

$$= 4V$$

$$V_{R_2} = I \cdot R_2 = \frac{0.2 \times 40}{10}$$

$$= 8V$$

Que. 2/-



classmate

Determine the current $I \rightarrow$

1^{st} - To first calculate the eqⁿ resistance of one ckt.

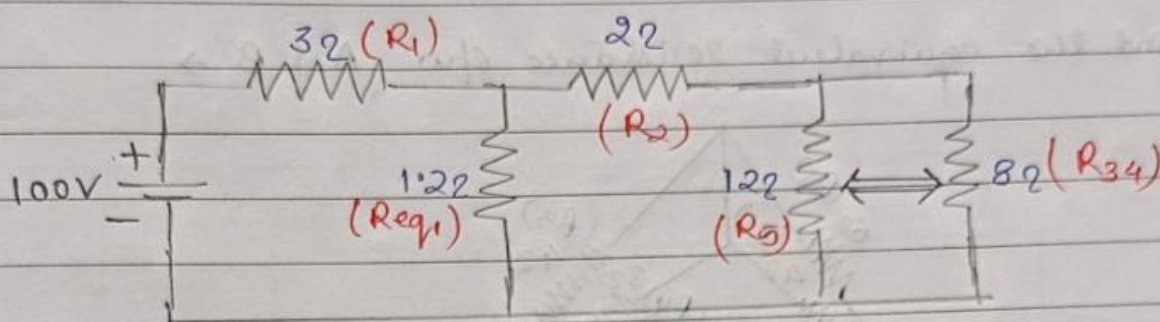
$$R_3 \text{ \& } R_4 - \text{series} \rightarrow R_3 + R_4 = 2 + 6 = 8\Omega = R_{34}$$

$$R_6 \parallel R_7 \parallel R_8 \Rightarrow \frac{1}{R_{eq1}} = \frac{1}{R_6} + \frac{1}{R_7} + \frac{1}{R_8}$$

$$\frac{1}{R_{eq1}} = \frac{1}{2} + \frac{1}{6} + \frac{1}{6}$$

$$= \frac{3+1+1}{6}$$

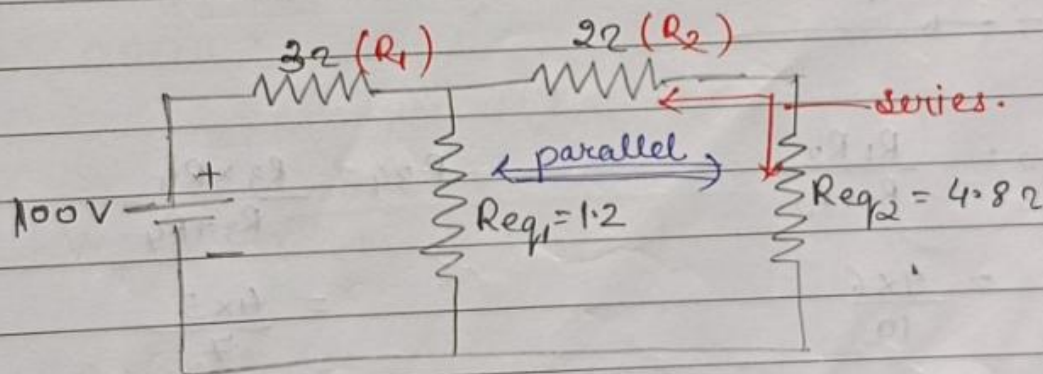
$$R_{eq1} = \frac{6}{5} = 1.2\Omega$$



$R_5 \text{ \& } R_{34} - \text{are in parallel}$

$\therefore$

$$R_{eq2} = \frac{R_5 \times R_{34}}{R_5 + R_{34}} = \frac{12 \times 8}{12 + 8} = \frac{12 \times 8^2}{20 \times 5} = \frac{24}{5} = 4.8\Omega$$

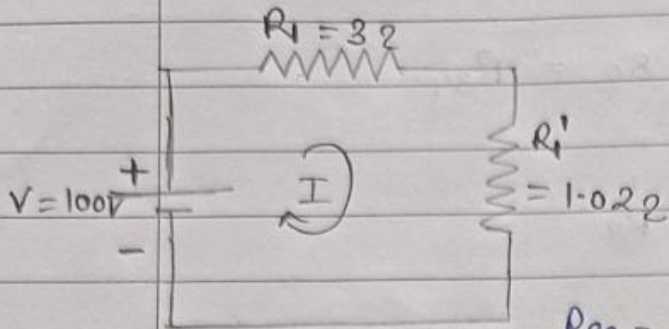


$R_2 \text{ \& } R_{eq2} - \text{are in series with each other, and in parallel with } R_{eq1}$

$$R_2' = R_2 + R_{eq2} \quad \& \quad R_2' \parallel R_{eq1}$$
$$= 2 + 4.8$$
$$= \underline{\underline{6.8 \Omega}}$$

$$R_1' = \frac{R_2' \times R_{eq1}}{R_2' + R_{eq1}}$$

$$= \frac{6.8 \times 1.2}{6.8 + 1.2} = \frac{8.16}{8} = 1.02$$

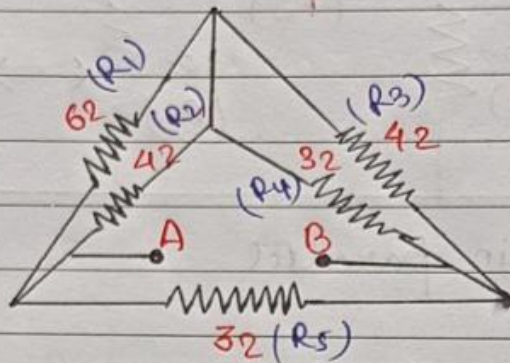


$$R_{eq} = R_1 + R_1' = 3 + 1.02 = \underline{\underline{4.02 \Omega}}$$

$$V = I R_{eq}$$

$$I = \frac{100}{R_{eq}} = \frac{100}{4.02} = \underline{\underline{24.87 \text{ Amp}}}$$

Ques/- find the equivalent resistance s/w A & B →



Solⁿ $R_1 \& R_2$ — are parallel

$R_3 \& R_4$ — — — — —

$$R_{12} = \frac{R_1 R_2}{R_1 + R_2}$$

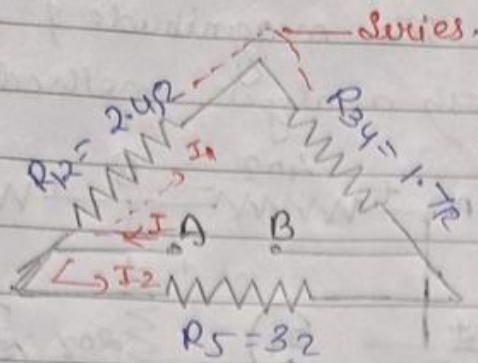
$$= \frac{4 \times 6}{10}$$

$$= \frac{24}{10} = 2.42$$

$$R_{34} = \frac{R_3 \times R_4}{R_3 + R_4}$$

$$= \frac{4 \times 3}{7}$$

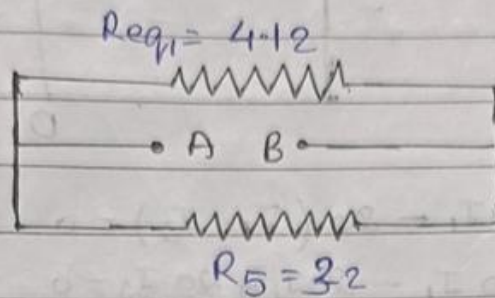
$$= \frac{12}{7} = 1.712$$



$$R_{eq1} =$$

$$R_2 + R_3 = 2.4 + 1.7$$

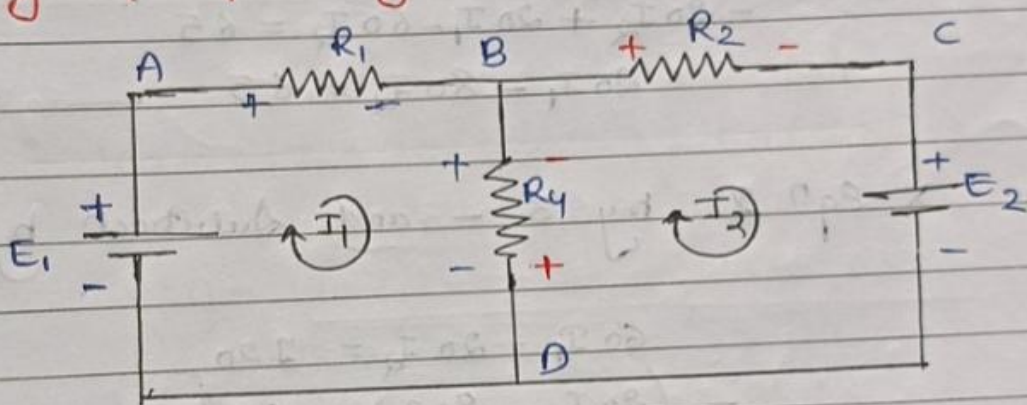
$$= \underline{\underline{4.1}}$$



$$R_{eq1} \parallel R_5 =$$

$$R_{eq} = \frac{R_{eq1} \times R_5}{R_{eq1} + R_5} = \frac{4.1 \times 3}{4.1 + 3} = \frac{12.3}{7.1} = \underline{\underline{1.732\Omega}}$$

Mesh Analysis (Loop Analysis)



Mesh ABDA.

$$E_1 - I_1 R_1 - (I_1 - I_2) R_4 = 0$$

$$E_1 = I_1 R_1 + I_1 R_4 - I_2 R_2$$

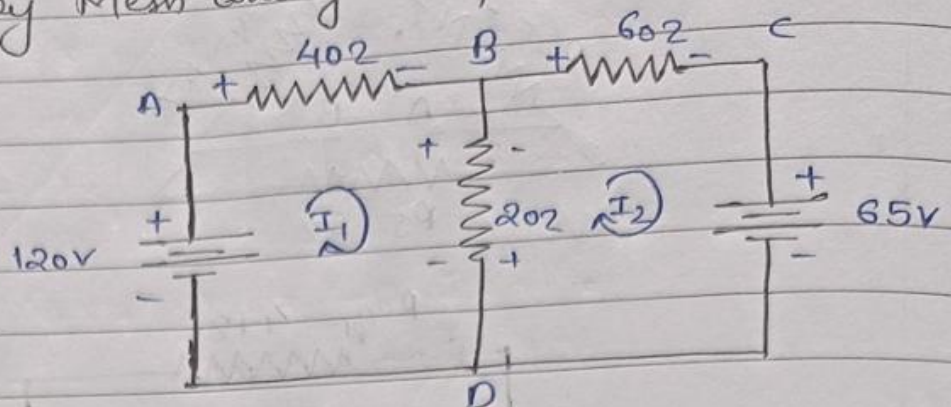
$$E_1 = I_1 (R_1 + R_4) - I_2 R_2$$

Mesh B.C.D.B.

$$-(I_2 - I_1) R_4 - I_2 R_2 - E_2 = 0$$

$$E_2 = -I_2 R_4 + I_1 R_4 - I_2 R_2$$

Q.1- In the given fig, find the magnitude & dirⁿ of each branch current by Mesh analysis method.



Solⁿ - Mesh 1 - ABDA.

$$120 - 40I_1 - 20(I_1 - I_2) = 0$$

$$120 - 40I_1 - 20I_1 + 20I_2 = 0$$

$$120 - 60I_1 + 20I_2 = 0$$

$$60I_1 - 20I_2 = 120 \quad \text{--- Eqⁿ (i)}$$

Mesh. 2 - BCDB.

$$-(I_2 - I_1)20 - 60I_2 - 65 = 0$$

$$-20I_2 + 20I_1 - 60I_2 = 65$$

$$20I_1 - 80I_2 = 65 \quad \text{--- Eqⁿ (ii)}$$

x Eqⁿ 2 by 3 - and Subtract by Eqⁿ (i)

$$60I_1 - 20I_2 = 120$$

$$-(20I_1 - 80I_2) = 65 \times 3$$

$$\Rightarrow 60I_1 - 20I_2 = 120$$

$$-60I_1 + 240I_2 = -195$$

$$220I_2 = -75$$

$$(I_2 = -0.3409) \text{ Amp.}$$

The '-' sign of I_2 indicates that the true dirⁿ of I_2 is anti-clockwise dirⁿ.

$$60 I_1 = 120 + 20 \times (-3.409)$$

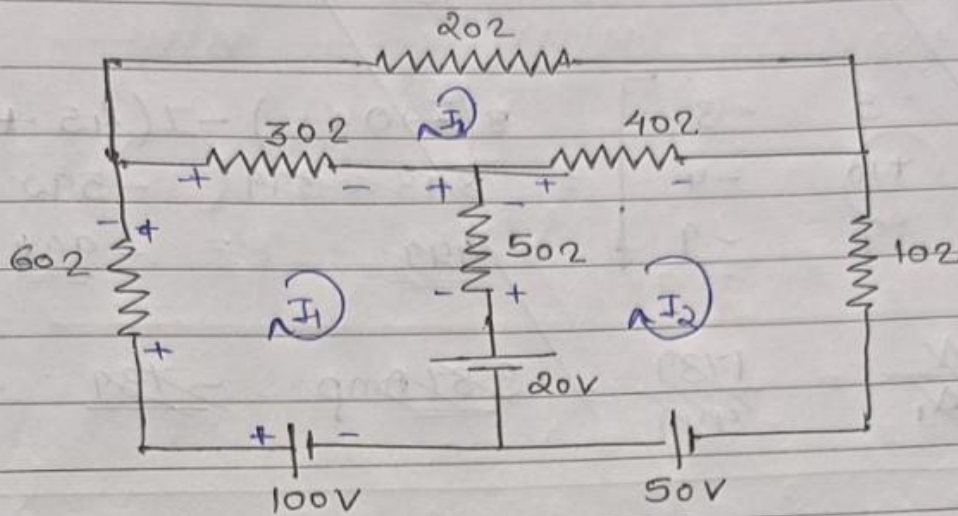
$$= 120 - 6.818$$

$$I_1 = \underline{1.886 \text{ Amp}}$$

Current through BD is -

$$I_1 + I_2 = \underline{2.27 \text{ Amp}}$$

7.2. Using mesh current method, calculate the current in each branch.



Soln) Mesh-1 $\rightarrow$

$$100 - 60 I_1 - 30(I_1 - I_3) - 50(I_1 - I_2) - 20 = 0$$

$$100 - 60 I_1 - 30 I_1 + 30 I_3 - 50 I_1 + 50 I_2 - 20 = 0$$

$$80 - 140 I_1 + 50 I_2 + 30 I_3 = 0$$

$$140 I_1 - 50 I_2 - 30 I_3 = 80 \quad \text{--- (i)}$$

$$14 I_1 - 5 I_2 - 3 I_3 = 8$$

Mesh-2 $\rightarrow$ $-50(I_2 - I_1) - 40(I_2 - I_3) - 10 I_2 + 50 + 20 = 0$

$$-50 I_2 + 50 I_1 - 40 I_2 + 40 I_3 - 10 I_2 + 70 = 0$$

$$50 I_1 - 100 I_2 + 40 I_3 = -70 \quad \text{--- (ii)}$$

$$-5 I_1 + 10 I_2 - 4 I_3 = +7$$

Mesh-3 $\rightarrow$ $-30(I_3 - I_1) - 20 I_3 - 40(I_3 - I_2) = 0$

$$-30 I_3 + 30 I_1 - 20 I_3 - 40 I_3 + 40 I_2 = 0$$

$$30 I_1 + 40 I_2 - 90 I_3 = 0$$

$$3 I_1 + 4 I_2 - 9 I_3 = 0$$

$$[R][I] = [E]$$

$$\begin{bmatrix} 14 & -5 & -3 \\ -5 & +10 & -4 \\ 3 & 4 & -9 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \\ 0 \end{bmatrix}$$

$$\Delta = 14(-90+16) - (5)(45+12) - 3(-20+30) =$$

$$= 1484 + 285 - 30 = -1036 + 285 + 150$$

$$= 1739 \quad = \underline{-736}$$

$$\Delta_1 = \begin{bmatrix} 8 & -5 & -3 \\ 7 & +10 & -4 \\ 0 & 4 & -9 \end{bmatrix} = 8(-90+16) - 7(45+12) + 0$$

$$= 848 - 399 - 592 - 399$$

$$= \underline{449} \quad = \underline{-999}$$

$$I_1 = \frac{\Delta}{\Delta_1} = \frac{1739}{449} = \underline{3.87 \text{ Amp}} \quad \frac{\Delta_1}{\Delta} = \underline{1.3487}$$

$$14x \quad (14I_1 - 5I_2 - 3I_3 = 8) \times 2 \quad \text{--- Eqn (i) } \times 2$$

$$28I_1 - 10I_2 - 6I_3 = 8$$

$$-5I_1 + 10I_2 - 4I_3 = 7$$

$$23I_1 - 10I_3 = 15 \quad \text{--- Eqn (iv)}$$

$$(-5I_1 + 10I_2 - 4I_3 = 7) \times 4$$

$$-20I_1 + 40I_2 - 16I_3 = 28$$

$$(3I_1 + 4I_2 - 9I_3 = 0) \times 10$$

$$30I_1 + 40I_2 - 90I_3 = 0$$

$$-20I_1 + 40I_2 - 16I_3 = 28$$

$$30I_1 + 40I_2 - 90I_3 = 0$$

$$-50I_1 + 74I_3 = 28$$

$$-25I_1 + 32I_3 = 14$$

$$(23I_1 - 10I_3 = 15) \times 32$$

$$736I_1 - 320I_3 = 480$$

$$(-25I_1 + 32I_3 = 14) \times 10$$

$$-250I_1 + 320I_3 = 140$$

$$736I_1 - 320I_3 = 480$$

$$-250I_1 + 320I_3 = 140$$

$$480I_1 = 620$$

$$I_1 = \underline{\underline{1.2917 \text{ Amp}}}$$

In Eqⁿ (iv) —

$$23 \times 1.2917 - 10 \times I_3 = 15$$

$$10I_3 = 14.71$$

$$I_3 = \underline{\underline{1.471 \text{ Amp}}}$$

$$3 \times 1.2917 + 4I_2 - 9 \times 1.471 = 0$$

$$3.87 + 4I_2 - 13.239 = 0$$

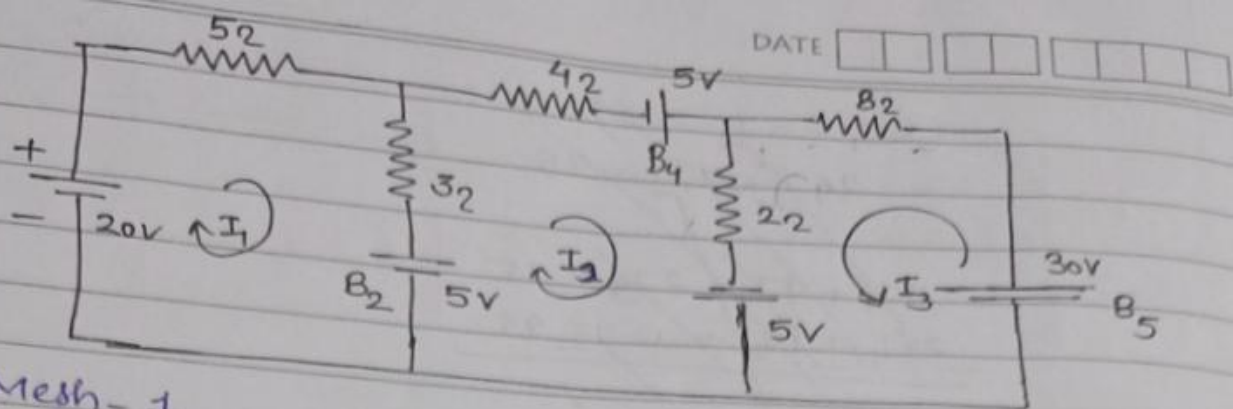
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∴ Current through 60Ω ——— $I_1 = 1.297 \text{ Amp.}$

————— 30Ω ——— $I_1 - I_3 = 1.297 - 1.471$

$= -0.174$ (current in
reverse dirⁿ.)



Solⁿ - Mesh-1

$$\begin{aligned}
 20 - 5I_1 - 3(I_1 - I_2) - 5 &= 0 \\
 15 - 8I_1 + 3I_2 &= 0 \\
 8I_1 - 3I_2 &= 15 \quad \text{--- (i)}
 \end{aligned}$$

Mesh-2 -

$$\begin{aligned}
 5 - 3(I_2 - I_1) - 4I_2 + 5 - 2(I_2 + I_3) + 5 &= 0 \\
 5 - 3I_2 + 3I_1 - 4I_2 + 5 - 2I_2 - 2I_3 + 5 &= 0 \\
 3I_1 - 9I_2 - 2I_3 + 15 &= 0 \\
 -3I_1 + 9I_2 + 2I_3 &= 15 \quad \text{--- (ii)}
 \end{aligned}$$

Mesh. 3 -

$$\begin{aligned}
 +30 - 8(I_3) - 2(I_2 + I_3) + 5 &= 0 \\
 -2I_2 - 8I_3 - 2I_3 + 35 &= 0 \\
 -2I_2 - 10I_3 &= -35 \\
 2I_2 + 10I_3 &= 35 \quad \text{--- (iii)}
 \end{aligned}$$

$$\begin{bmatrix} 8 & -3 & 0 \\ -3 & 9 & 2 \\ 0 & 2 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 15 \\ 35 \end{bmatrix}$$

$$\begin{aligned}
 \Delta &= 8(90 - 4) + 3(-30) \\
 &= 688 + (-90) \\
 &= \underline{\underline{598}}
 \end{aligned}$$

$$\Delta_1 = \begin{bmatrix} 15 & -3 & 0 \\ 15 & 9 & 2 \\ 35 & 2 & 10 \end{bmatrix} \Rightarrow 15(90 - 4) + 3(70 - 150) = 1290 - (-280) = 1530$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{1530}{598} = \underline{2.55 \text{ Amp}}$$

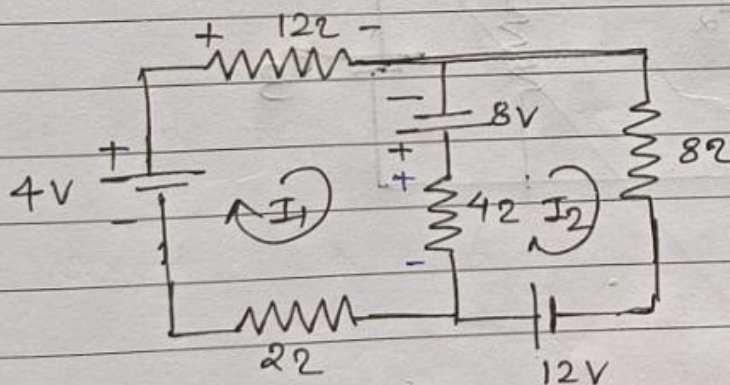
$$\Delta_2 = \begin{bmatrix} 8 & 15 & 0 \\ -3 & 15 & 2 \\ 0 & 35 & 10 \end{bmatrix} = 8(150 - 70) - 15(30) + 0 = 640 + 450 = \underline{1090}$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{1090}{598} = \underline{1.823 \text{ Amp}}$$

$$\Delta_3 = \begin{bmatrix} 8 & -3 & 15 \\ -3 & 9 & 15 \\ 0 & 2 & 35 \end{bmatrix} = 8(315 - 30) + 3(-105) + 15(-6) = 2280 - 315 - 90 = \underline{1875}$$

$$I_3 = \frac{\Delta_3}{\Delta} = \frac{1875}{598} = \underline{3.135 \text{ Amps}}$$

Ques-



Determine all the loop currents.

Mesh-1 -

$$4 - 12(I_1) + 8 - 4(I_1 - I_2) - 2I_1 = 0$$

$$-18I_1 + 4I_2 + 12 = 0$$

$$9I_1 - 4I_2 = 6 \quad \text{--- (ii)}$$

Mesh-2 -

$$-8 - 8I_2 + 12 - 4(I_2 - I_1) = 0$$

$$4 - 8I_2 + 4I_1 = 0$$

$$I_1 - 3I_2 + 1 = 0$$

$$I_1 - 3I_2 = -1 \quad \text{--- (ii)} \quad \times 9.$$

~~$$9I_1 - 4I_2 = 6$$~~

~~$$9I_1 - 27I_2 = -9$$~~

~~$$+ \quad +$$~~

$$23I_2 = 14$$

$$I_2 = \frac{14}{23} = 0.607$$

$$\approx \underline{\underline{0.61 \text{ Amp.}}}$$

$$I_1 = -1 + 3 \times 0.61$$

$$= \underline{\underline{0.83 \text{ Amp}}}$$

Superposition Theorem -

The Superposition Theorem states that -

"In a linear d.c. n/w containing more than one energy source, the resultant potential difference across or current through any element is equal to the algebraic sum of potential differences or currents for that element produced by each source acting alone with all other independent s.c.s. —

Ideal voltage source $\xrightarrow{\text{replaced by}}$ short ckt
 Ideal current source $\xrightarrow{\text{replaced by}}$ open circuit

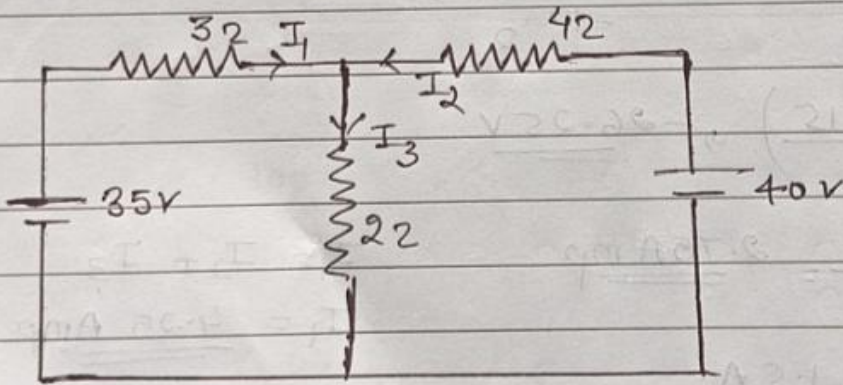
ideal voltage source replaced by open circuit.

9 steps —

- 1) Select any one energy source
- 2) replace all other energy source by :-

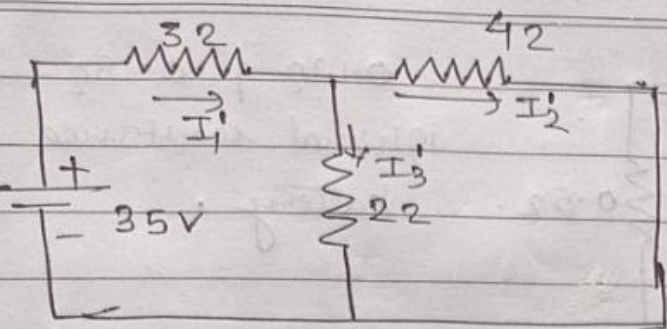
- their internal series resistance for voltage source
- - - - - short - - - - - current source

- 3) With only one energy source calculate the voltage drops or branch current.
- 4) Repeat the above steps for each sources individually
- 5) Add algebraically the one voltage across or current obtained due to individual source to obtain the combined effect of all the sources.



find the diff
branch current one.
super-position theorem.

→ Replace 40V battery by short ckt.



$$2\Omega \parallel 4\Omega \Rightarrow \frac{2 \times 4}{2+4} = \frac{8}{6} = \frac{4}{3}$$

$$\Rightarrow 1.33\Omega$$

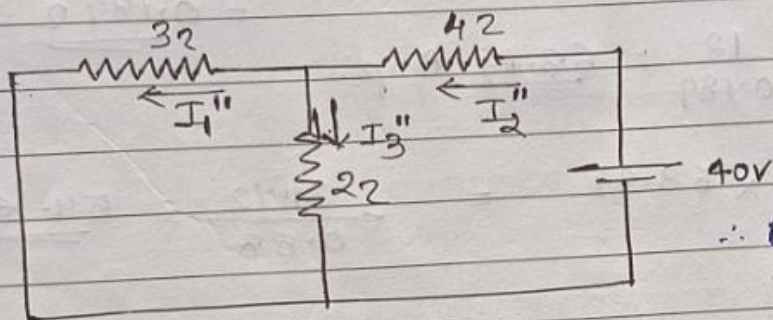
$$R_{eq} = 3 + 1.33 = 4.33\Omega$$

$$I_1' = \frac{V}{R} = \frac{35}{4.33} = 8.08 \text{ Amp}$$

$$I_1' = 8.08 \text{ Amp} \quad I_2' = \frac{2}{4+2} \times 8.08 = \frac{2}{6} \times 8.08 = 2.69 \text{ Amp}$$

$$I_1' = I_2' + I_3' \Rightarrow I_3' = 8.08 - 2.69 = 5.39 \text{ Amp}$$

Now replace 35V battery by the short circuit :-



$$3\Omega \parallel 2\Omega \Rightarrow$$

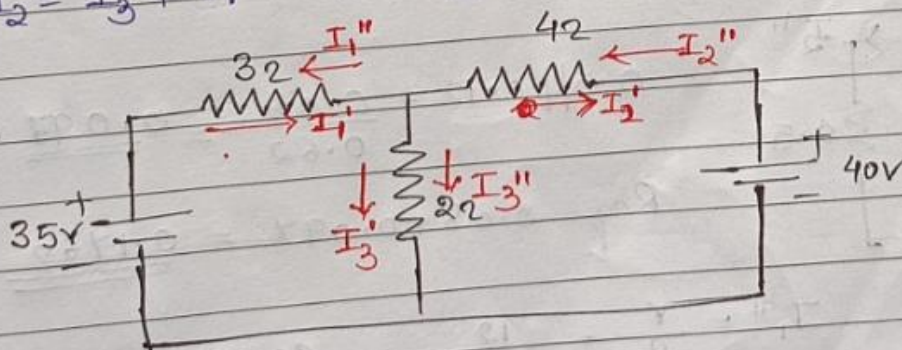
$$\frac{3 \times 2}{3+2} = \frac{6}{5} = 1.2 \text{ Amp}$$

$$\therefore R_{eq} = 4 + 1.2 = 5.2 \text{ Amp}$$

$$\therefore I_2'' = \frac{V}{R_{eq}} = \frac{40}{5.2} = 7.69 \text{ Amp}$$

$$I_1'' = \frac{2}{3+2} \times 7.69 = 3.07$$

$$I_2'' = I_3'' + I_1'' \Rightarrow I_3'' = 7.69 - 3.07 = 4.62 \text{ Amp}$$

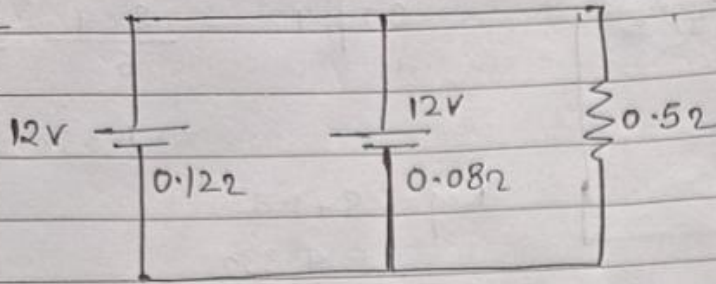


$$I_1 = I_1' - I_1'' = 8.08 - 3.07 = 5.01 \text{ Amp}$$

$$I_2 = I_2' - I_2'' = 2.69 - 7.69 = -5 \text{ Amp}$$

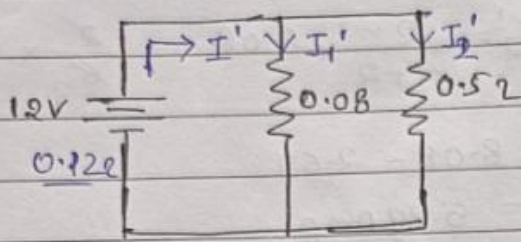
$$I_3 = I_3' + I_3'' = 5.39 + 4.62 = 10.01$$

Que. 2 -



0.12Ω & 0.08Ω are internal resistance of the battery.

→ Replace — 12V battery with internal resistance — 0.08Ω



$$R_{eq} = \frac{0.5 \times 0.08}{0.5 + 0.08} = \frac{0.04}{0.58} = 0.069\Omega$$

$$I' = \frac{V}{R_{eq}} = \frac{12}{0.069} = 173.91 \text{ Amp}$$

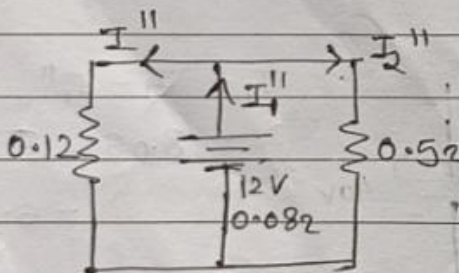
$$R_t = \text{total ckt resistance} = 0.12 + 0.069 = 0.189\Omega$$

$$I' = \frac{V}{R_{eq}} = \frac{12}{0.189} = 63.49 \text{ Amp.}$$

$$I_2' = \frac{0.5}{0.08 + 0.5} \times 63.49 = \frac{31.72}{0.58} = 54.69 \text{ Amp.}$$

$$\begin{aligned} I_2' &= I' - I_1' \\ &= 63.49 - 54.69 \\ &= 8.8 \text{ Amp.} \end{aligned}$$

→ Now in next step replace 12V battery with its internal resistance — 0.12Ω



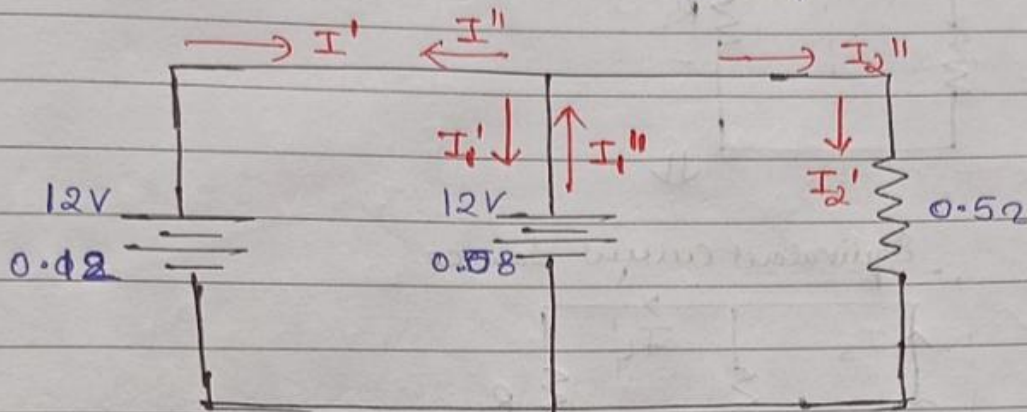
$$0.5 \parallel 0.12 = \frac{0.06}{0.62} = 0.097\Omega$$

$$R_{eq} = 0.08 + 0.097 = 0.176\Omega$$

$$I_1'' = \frac{V}{R_{eq}} = \frac{12}{0.176} = 68.18 \text{ Amp.}$$

$$I_2'' = \frac{0.5}{0.5 + 0.12} \times 68.18 = \frac{34.09}{0.62} = \underline{54.98 \text{ Amp.}}$$

$$\begin{aligned} I_1'' &= I'' + I_2'' \Rightarrow I'' = I_1'' - I_2'' \\ I_2' &= I_1'' - I'' = 68.18 - 54.98 \\ &= 68.18 - 54.98 = \underline{14 \text{ Amp.}} = \underline{14 \text{ Amp.}} \end{aligned}$$



~~$$I = I' - I'' = 63.49 - 14 = 49.49 \text{ Amp.}$$~~

~~$$I_1 = I_1' - I_1'' = 54.69 - 68.18 = -13.49 \text{ Amp.}$$~~

~~$$I_2 = I_2' + I_2'' = 8.8 + 54.98 =$$~~

$$I = I' - I'' = 63.49 - 54.98 = 8.51 \text{ Amp.}$$

$$I_1 = I_1' - I_1'' = 54.69 - 68.18 = -13.49 \text{ Amp.}$$

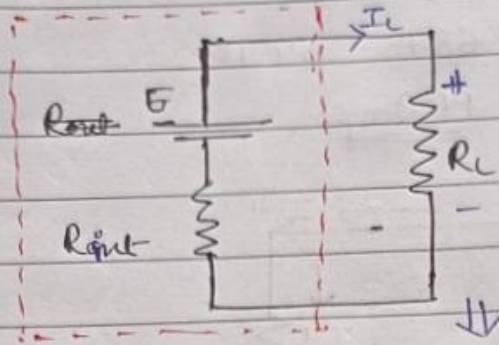
$$I_2 = I_2' + I_2'' = 8.8 + 14 = \underline{22.8 \text{ Amp}}$$

Source Conversion $\rightarrow$

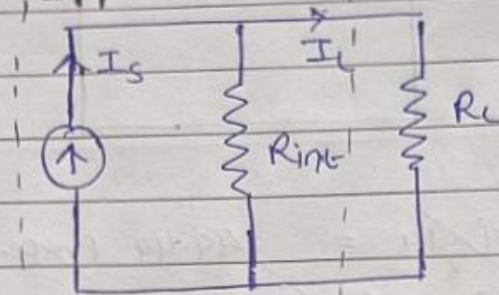
$\rightarrow$ Voltage to Current Source

A real voltage source can be converted to real voltage current source.

R_{int} internal resistance of one battery.



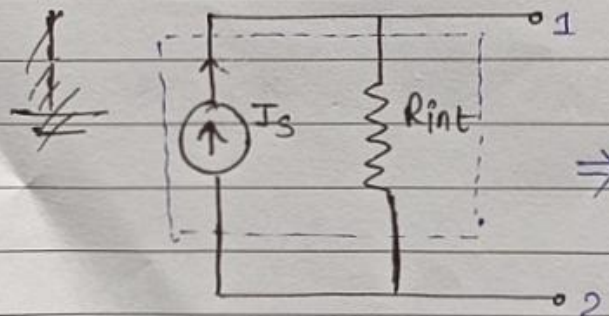
Equivalent Current Source



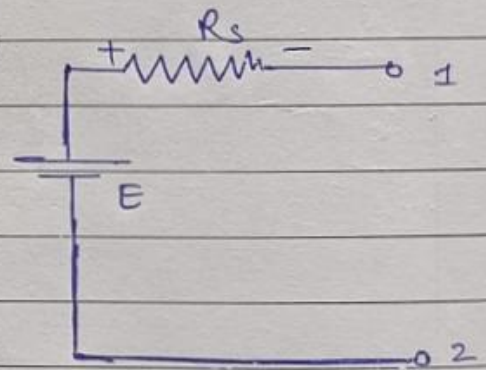
$$I_L = \frac{E}{R_{int} + R_L} = \frac{\frac{E}{R_{int}}}{\frac{R_{int} + R_L}{R_{int}}} = \frac{E}{R_{int}} \times \frac{R_{int}}{R_{int} + R_L}$$

$$I_L = I_s \times \left(\frac{R_{int}}{R_{int} + R_L} \right)$$

$\rightarrow$ Current to Voltage Source Conversion



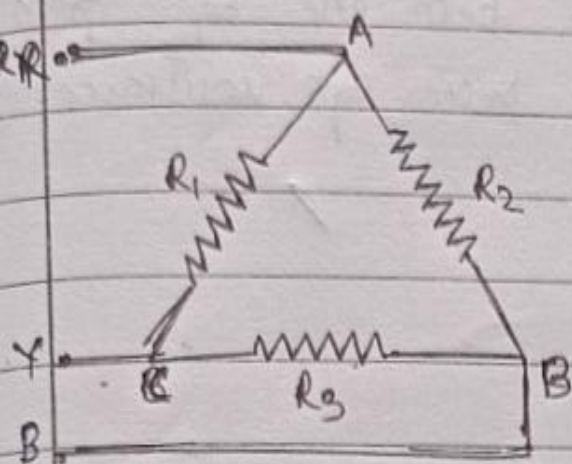
Real Current Source.



Eqⁿ Voltage Source.

Delta Network —

Consider the following Delta Network —



$$\begin{aligned} R_{CA} &= R_1 \parallel (R_2 + R_3) \\ &= \frac{R_1 (R_2 + R_3)}{R_1 + R_2 + R_3} \end{aligned}$$

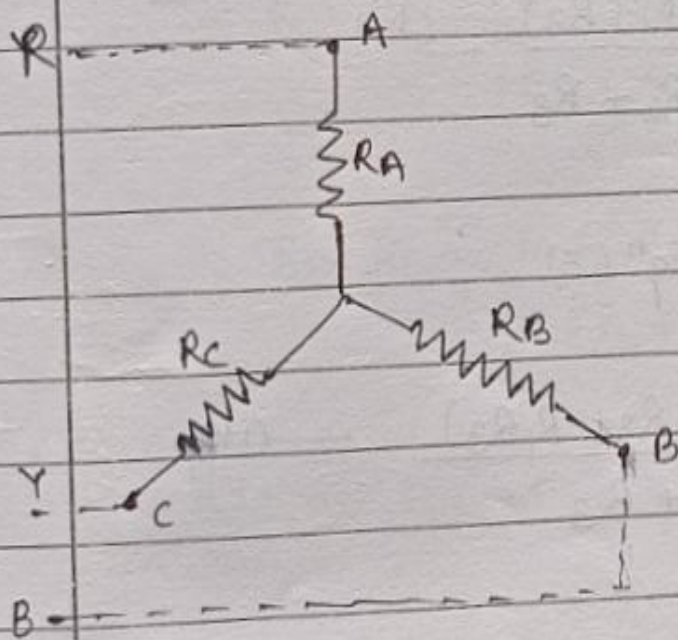
$$\begin{aligned} R_{BC} &= R_3 \parallel (R_1 + R_2) \\ &= \frac{R_3 (R_1 + R_2)}{R_1 + R_2 + R_3} \end{aligned}$$

$$\begin{aligned} R_{AB} &= R_2 \parallel (R_1 + R_3) \\ &= \frac{R_2 (R_1 + R_3)}{R_1 + R_2 + R_3} \end{aligned}$$

The following Eqⁿ represent the Eqⁿ resistance s/w terminal when the third terminal is kept open.

Star Network —

Consider the following Star Network —



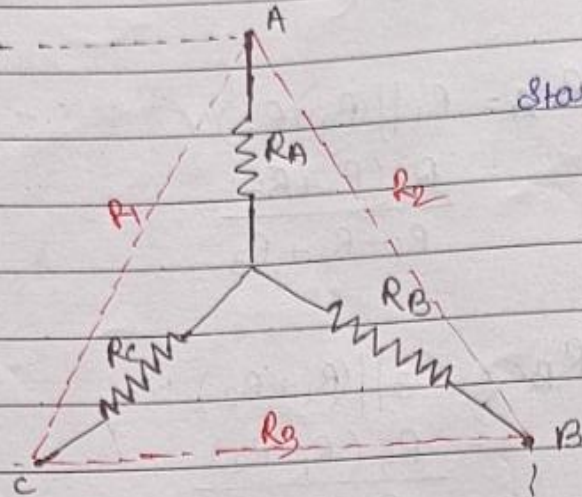
$$R_{AB} = R_A + R_B \text{ — (i)}$$

$$R_{BC} = R_B + R_C \text{ — (ii)}$$

$$R_{CA} = R_C + R_A \text{ — (ii)}$$

Star to Delta conversion:- Delta to star conversion:-

compare both one Eqⁿ from star and Delta Eqⁿ resistance.



$$R_{AB} = R_A + R_B = \frac{R_2(R_1 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- (i)}$$

$$R_B + R_C = \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3} \quad \text{--- (ii)}$$

$$R_A + R_C = \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- (iii)}$$

Add above Eqⁿ —

$$\text{Eqⁿ (i) + Eqⁿ (2) + Eqⁿ (3)}$$

$$2(R_A + R_B + R_C) = \frac{2(R_1R_2 + R_2R_3 + R_1R_3)}{R_1 + R_2 + R_3} \quad \text{--- (iv)}$$

Subtract Eqⁿ (ii) from Eqⁿ (iv)

$$R_A + \cancel{R_B} + \cancel{R_C} = \frac{R_1R_2 + \cancel{R_2R_3} + \cancel{R_1R_3}}{R_1 + R_2 + R_3} - \left(\frac{\cancel{R_1R_3} + \cancel{R_2R_3}}{R_1 + R_2 + R_3} \right)$$

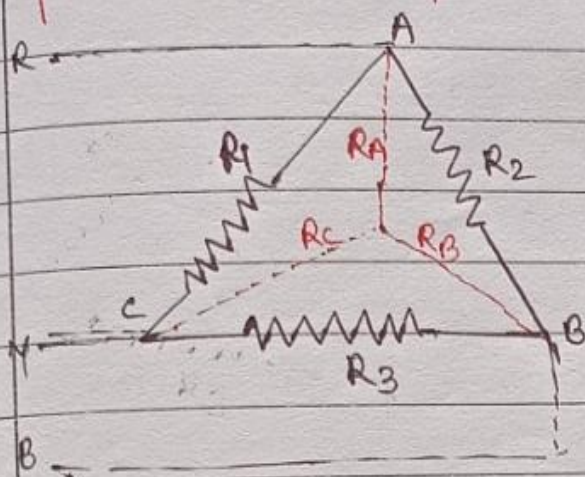
$$R_A = \frac{R_1R_2}{R_1 + R_2 + R_3} \quad \text{--- (v)}$$

Similarly -

$$R_B = \frac{R_2 R_3}{R_1 + R_2 + R_3} \quad \text{--- (vi)}$$

$$R_C = \frac{R_1 R_3}{R_1 + R_2 + R_3} \quad \text{--- (vii)}$$

Star to Delta Conversion ---



Now for the Star - Delta Conversion
Multiply the Star Eqn ---

Star Eqn --- (v), (vi), (vii)

Multiply any 2 sets & then add
all the three Eqn ---

Multiply Eqn (v) & (vi) --- then,

$$R_A \cdot R_B = \frac{R_1 R_2^2 R_3}{(R_1 + R_2 + R_3)^2}$$

$$R_A \cdot R_C = \frac{R_1^2 R_2 R_3}{(R_1 + R_2 + R_3)^2}$$

$$R_B \cdot R_C = \frac{R_1 R_2 R_3^2}{(R_1 + R_2 + R_3)^2}$$

Now add all the above Eqn ---

$$R_A R_B + R_B R_C + R_A R_C = \frac{R_1 R_2^2 R_3 + R_1 R_2 R_3^2 + R_1^2 R_2 R_3}{(R_1 + R_2 + R_3)^2}$$

$$R_A R_B + R_B R_C + R_A R_C = \frac{R_1 R_2 R_3 (R_2 + R_3 + R_1)}{(R_1 + R_2 + R_3)^2}$$

$$R_A R_B + R_B R_C + R_C R_A = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3}$$

Now, dividing Eqⁿ (viii) by Eqⁿ (vi)

$$\frac{R_A R_B + R_B R_C + R_C R_A}{R_B} = \frac{\frac{R_1 R_2 R_3}{R_1 + R_2 + R_3}}{\frac{R_2 R_3}{R_1 + R_2 + R_3}}$$

$$R_A + R_C + \frac{R_C R_A}{R_B} = R_1$$

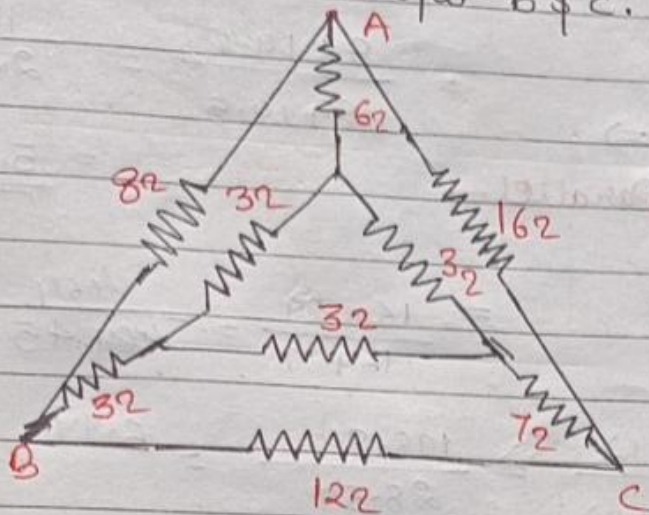
$$R_1 = R_A + R_C + \frac{R_A R_C}{R_B}$$

Similarly,

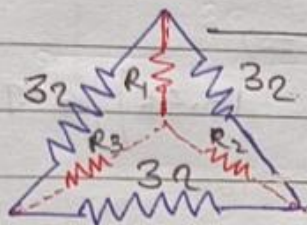
$$R_2 = R_A + R_B + \frac{R_A R_B}{R_C}$$

$$R_3 = R_B + R_C + \frac{R_B R_C}{R_A}$$

1- Find the resistance b/w B & C.



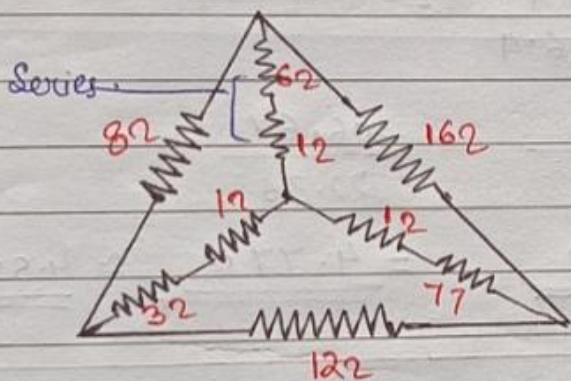
2- Convert the inner delta into star -



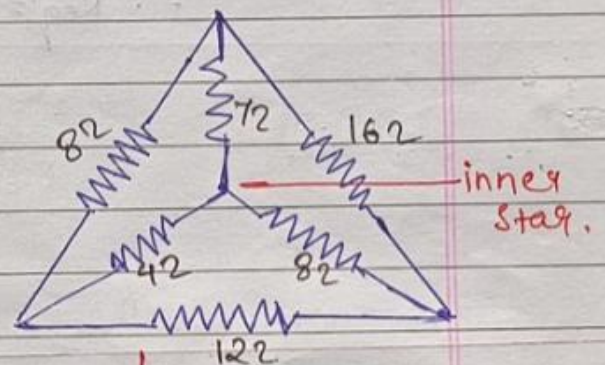
inner delta

$$R_1 = \frac{3 \times 3}{3+3+3} = \frac{9}{9} = 1\Omega$$

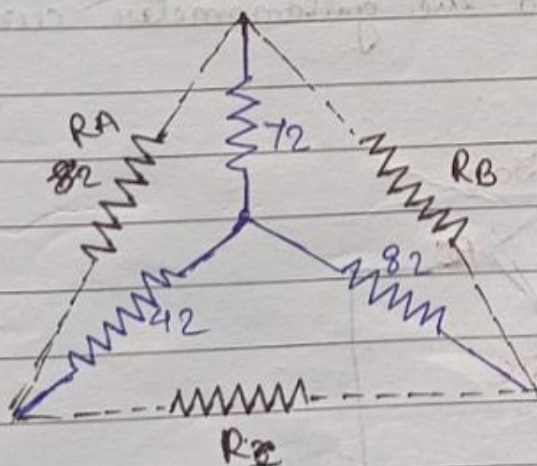
Similarity, $R_2 \text{ \& } R_3 = \frac{3 \times 3}{3+3+3} = \frac{9}{9} = 1\Omega$



=>



inner star
Now convert the star to Delta

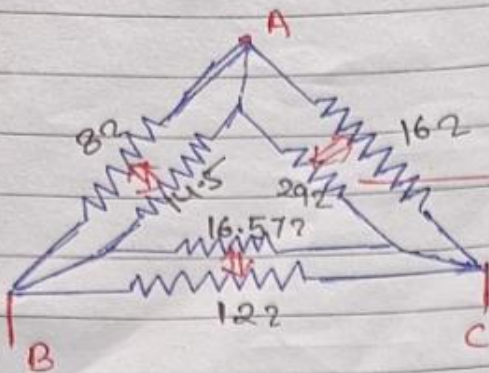


$$\therefore R_A = 7 + 4 + \frac{7 \times 4}{8} = \frac{29}{2} = 14.5\Omega$$

$$R_B = 7 + 8 + \frac{7 \times 8}{4} = 29\Omega$$

$$R_C = 4 + 8 + \frac{4 \times 8}{7}$$

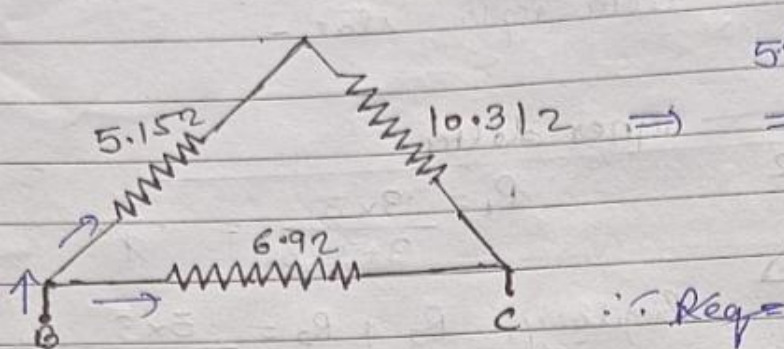
$$= 12 + \frac{32}{7} = 16.57\Omega$$



$$= \frac{8 \times 14.5}{8 + 14.5} = \frac{116}{22.5} = 5.152$$

$$= \frac{16 \times 29}{16 + 29} = \frac{464}{45} = 10.312$$

$$= \frac{16.57 \times 12}{16.57 + 12} = \frac{198.84}{28.57} = 6.94$$



$$5.152 + 10.312 = 15.462$$

$$15.462 \parallel 6.94$$

$$\therefore R_{eq} = \frac{15.462 \times 6.94}{15.462 + 6.94}$$

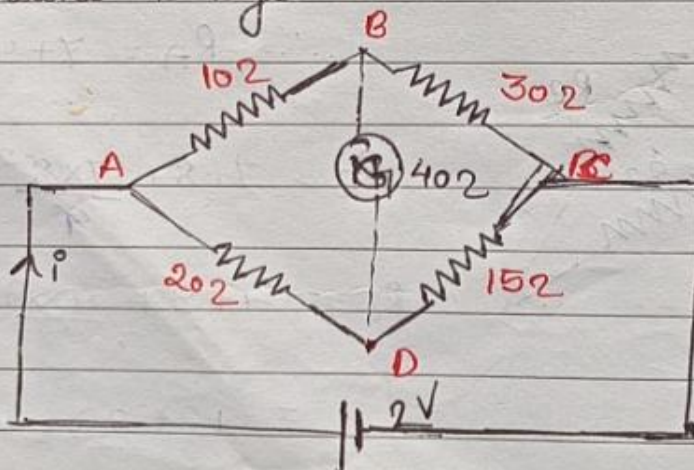
$$= \frac{106.67}{22.402}$$

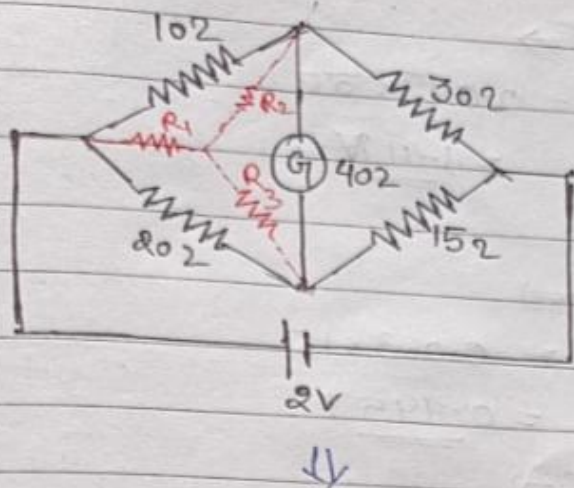
$$= \frac{106.67}{22.36}$$

$$= 4.772$$

$$= 4.772 \approx 4.82$$

Q. Using star/Delta transformation, find the galvanometer current in the wheatstone bridge.

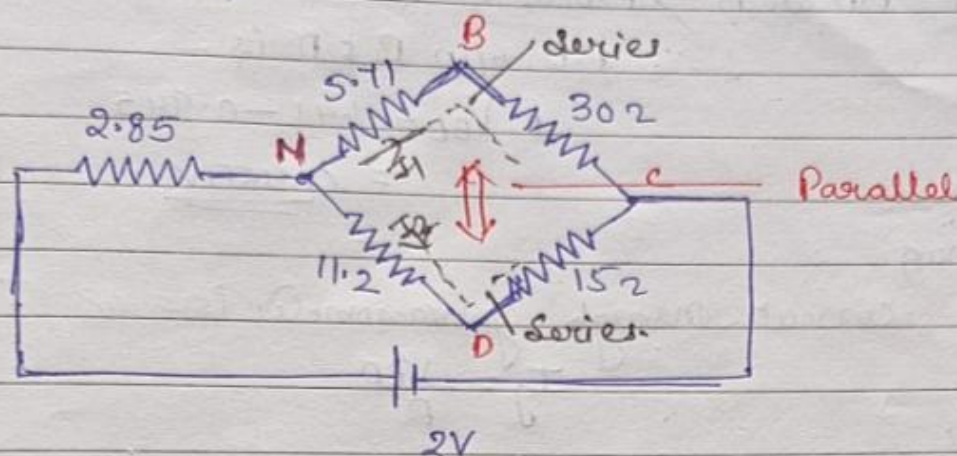




$$R_1 = \frac{10 \times 20}{10 + 20 + 40} = \frac{200}{70} = 2.852$$

$$R_2 = \frac{10 \times 40}{10 + 20 + 40} = \frac{400}{70} = 5.712$$

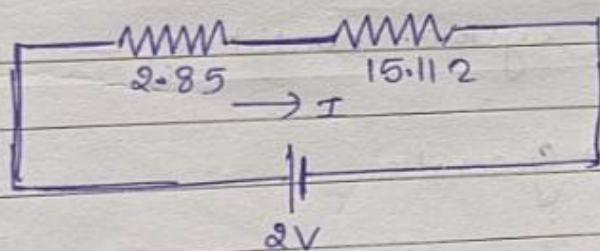
$$R_3 = \frac{40 \times 20}{10 + 20 + 40} = \frac{800}{70} = 11.42$$



$$\rightarrow 5.71 + 30 = 35.712$$

$$\rightarrow 11.2 + 15 = 26.22$$

$$R_{NC} = \frac{35.71 \times 26.2}{35.71 + 26.2} = \frac{935.60}{61.91} = 15.112$$



$$R_{eq} = 15.11 + 2.85 = 17.96$$

$$I = \frac{E}{R_{eq}} = \frac{2}{17.96} = 0.11 \text{ Amp}$$

At pt. N — current divides into two —

$$I_1 = \frac{26.2}{35.7 + 26.2} \times 0.11 = \frac{2.882}{61.9} = 0.047 \text{ Amp}$$

$$I_2 = \frac{35.71}{35.7 + 26.2} \times 0.11 = \frac{3.92}{61.91} = 0.063 \text{ Amp}$$

potential of B w.r.t. C $\Rightarrow$

$$I \times R = 0.047 \times 30 \\ = \underline{1.41 \text{ V}}$$

potential of D w.r.t. C $\Rightarrow$

$$I \times R = 0.063 \times 15 \\ = \underline{0.945 \text{ V}}$$

Hence, P.D at B $>$ P.D at C $\rightarrow$

$\therefore$ P.D b/w B & D is -

$$\text{P.D } V_{BD} = 1.41 - 0.945 \\ = \underline{0.465 \text{ V}}$$

thus -

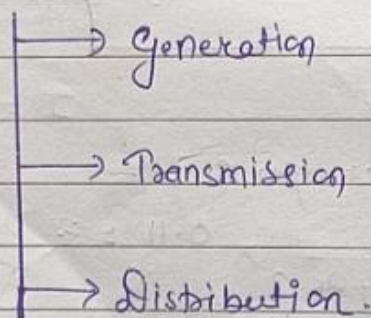
current through galvanometer is -

$$I_g = \frac{V_{BD}}{R}$$

$$= \frac{0.465}{40} = \underline{0.011 \text{ Amp.}}$$

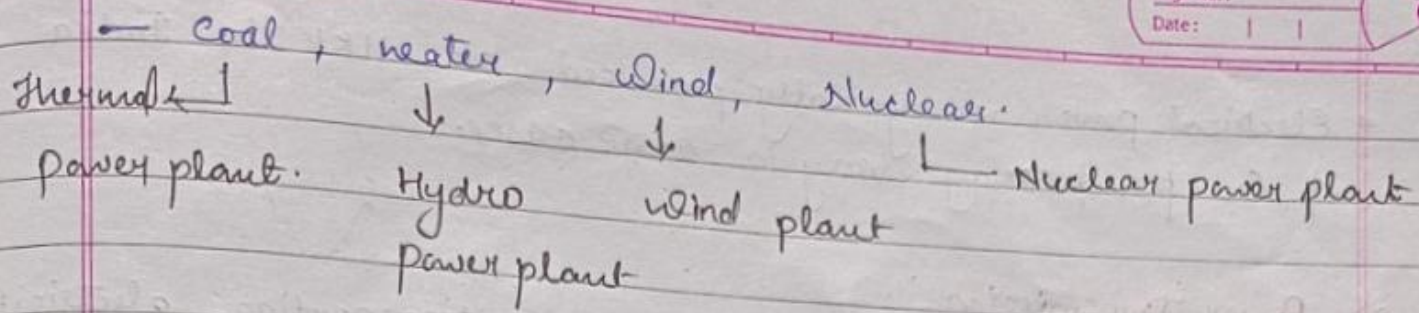
Q. Explain the block diagram of elementary power system showing the stages of - generation, transmission & distribution.

- An electrical power system consists of -



1) power Generation -

Power Generation is a process of producing electricity from various energy sources -



Generating Station

Step 6.6 KV, 11KV, 13.8 KV, 15.75KV

Primary transmission

Step-up X-former

(11/220 KV, 132 KV or 110 KV or 400 KV or 765 KV)

Secondary transmission distribution

Step down X-former

(132 or 220 KV / 33 KV or 66 KV)

Primary distribution

Step down X-former

(33 KV / 11 KV or 6.6 KV)

Big consumers

Large factories.

Secondary distribution

Step down X-former

(11 KV / 400 V — for 3- ϕ
or 230 KV — for 1- ϕ)

Service mains

Service mains

Consumers

- * Electrical power is typically produced at 11KV in India & Europe, & may vary in some cases.

2) Power Transmission -

Transmission is the process of transporting electricity over long distance from power plant

↓
through high power line.

Substation

- After generating - the voltage is step-up through the X-former to - 400KV or 765KV - at primary transmission level.

{ stepping-up voltage depends on the distance at which power is X-mitted }

[stepping-up voltage reduces I^2R loss,

i.e. when $V \uparrow$, one reduces by relative amount & the power remains constant, Hence it reduces the losses]

- Now, before distribution - again the voltage is stepped-down to - 33KV or 66KV, at receiving station.

3) Power Distribution -

- Secondary X-mission line emerges from this receiving station to connect to the near substation for distribution.
- the voltage again step-down $\rightarrow$ to 11KV or 6.6KV - and distributed to - large factories or big consumers $\rightarrow$ called as primary distribution.
- from this substation - feed emerges &

The feeders are either — Overhead x-lines
or Underground x-lines. Cables.

— Again & finally the voltage is stepped down.

↓
420V or 415 V — for 3- ϕ
230V — $\rightarrow$ 1- ϕ

and supplied to the consumers, the feeders, distributors or service mains.